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(54) **SEMICONDUCTOR DEVICES AND METHODS OF MANUFACTURE**

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(57)

ABSTRACT

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(60) Continuation of application No. 17/881,866, filed on Aug. 5, 2022, now Pat. No. 12,349,379, which is a division of application No. 16/872,058, filed on May 11, 2020, now Pat. No. 11,450,754.

(60) Provisional application No. 62/927,582, filed on Oct. 29, 2019.

Publication Classification

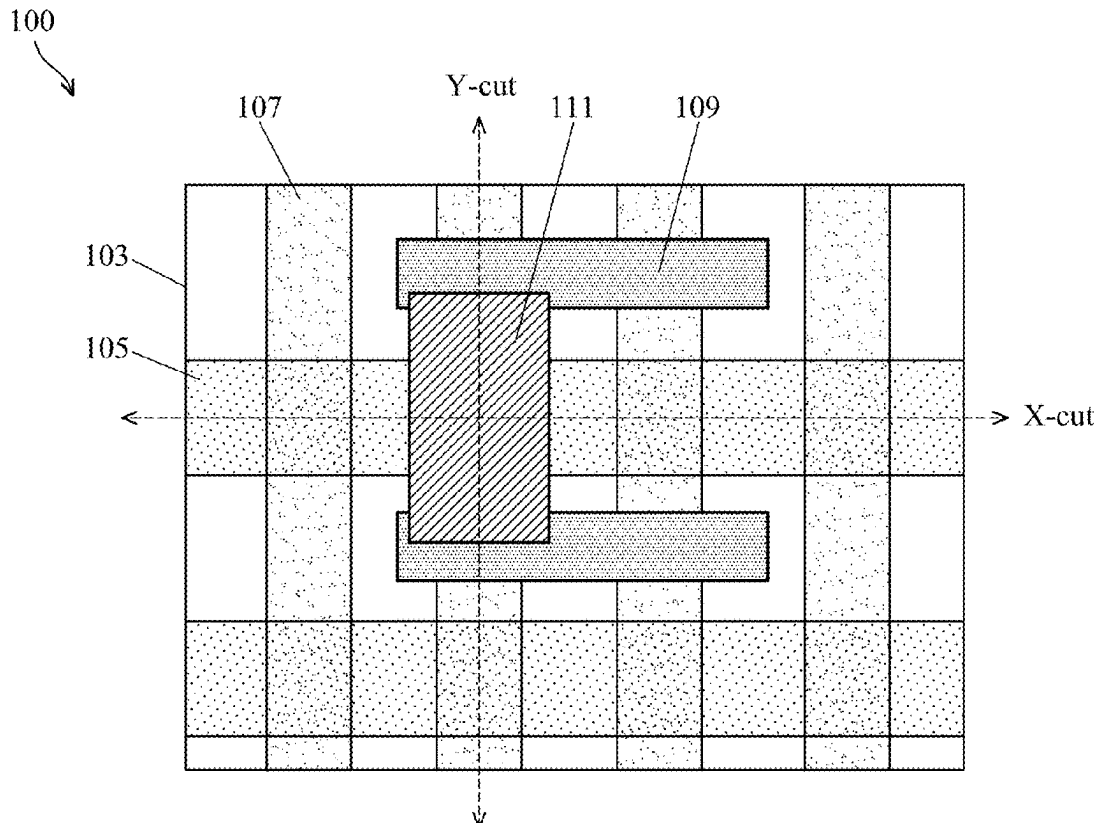
(51) **Int. Cl.**

H10D 30/01 (2025.01)

H10D 30/62 (2025.01)

H10D 30/67 (2025.01)

Semiconductor devices using a dielectric structure and methods of manufacturing are described herein. The semiconductor devices are directed towards gate-all-around (GAA) devices that are formed over a substrate and are isolated from one another by the dielectric structure. The dielectric structure is formed over the fin between two GAA devices and cuts a gate electrode that is formed over the fin into two separate gate electrodes. The two GAA devices are also formed with bottom spacers underlying source/drain regions of the GAA devices. The bottom spacers isolate the source/drain regions from the substrate. The dielectric structure is formed with a shallow bottom that is located above the bottoms of the bottom spacers.



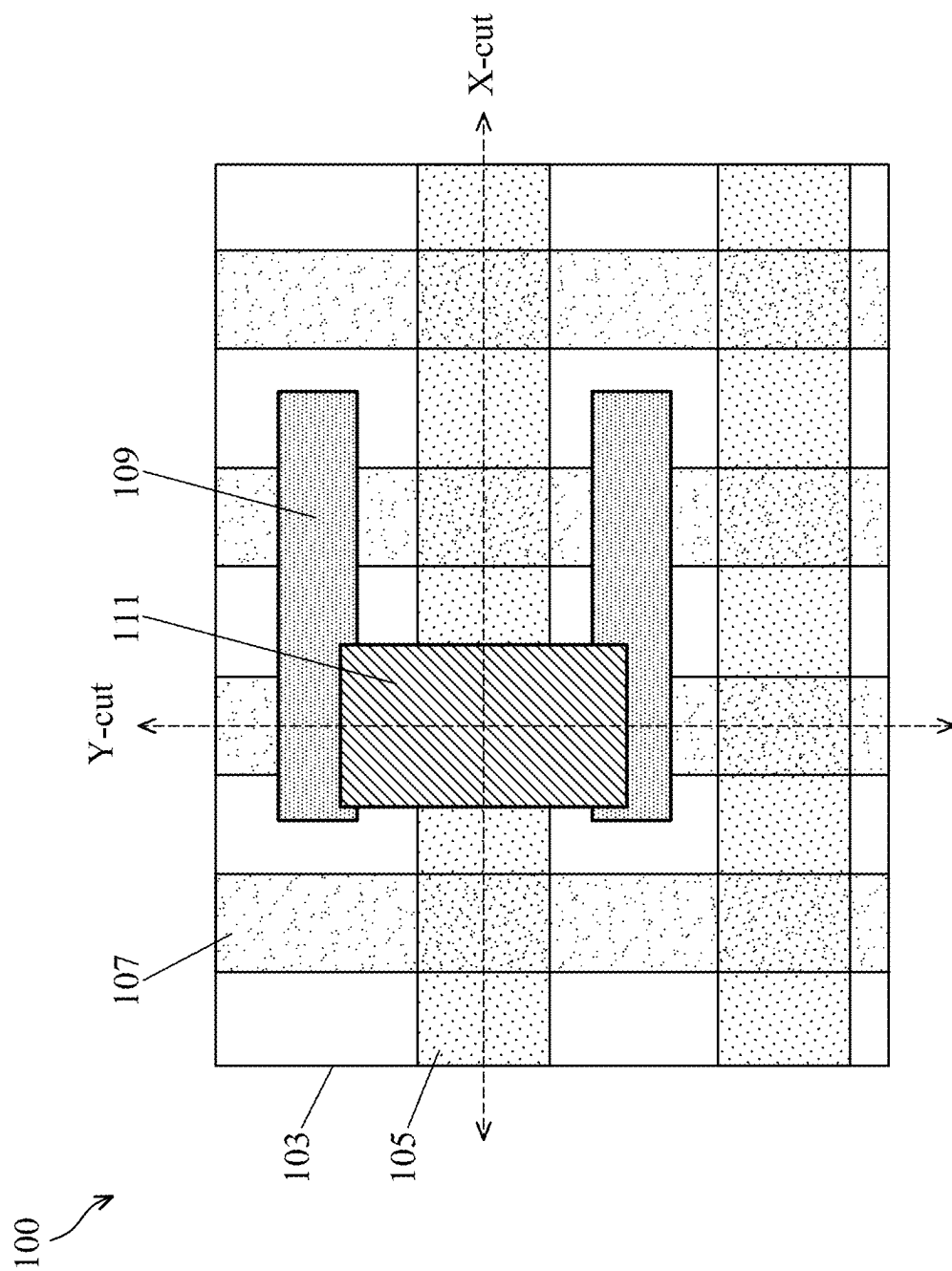
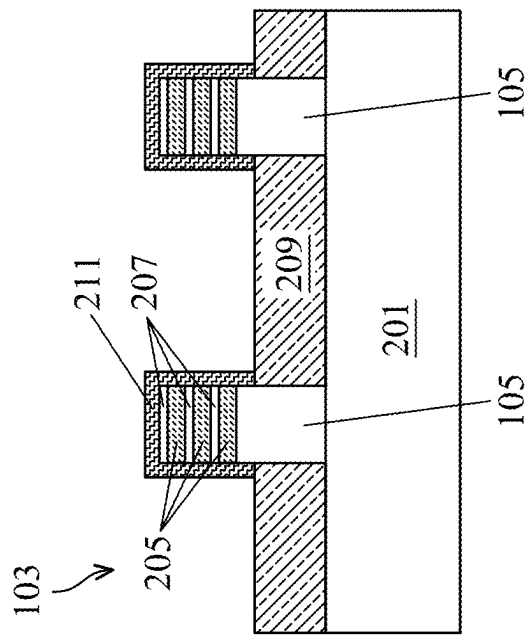
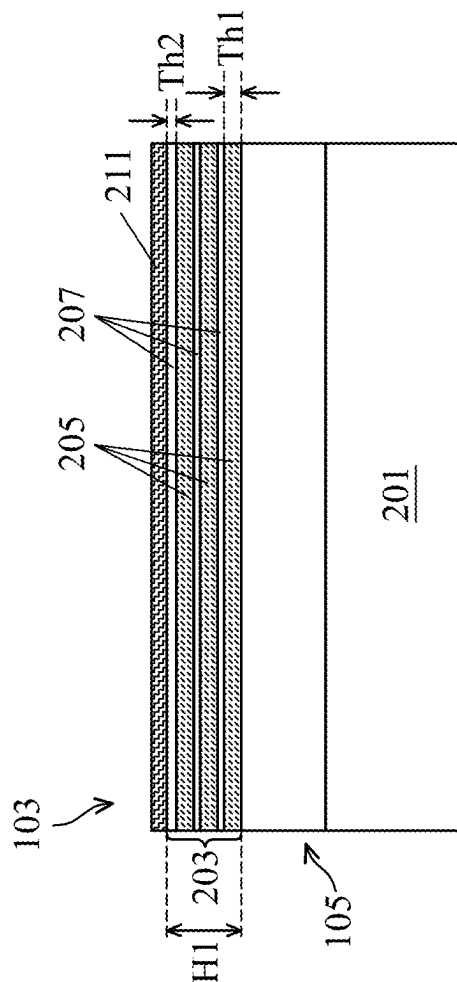


Figure 1



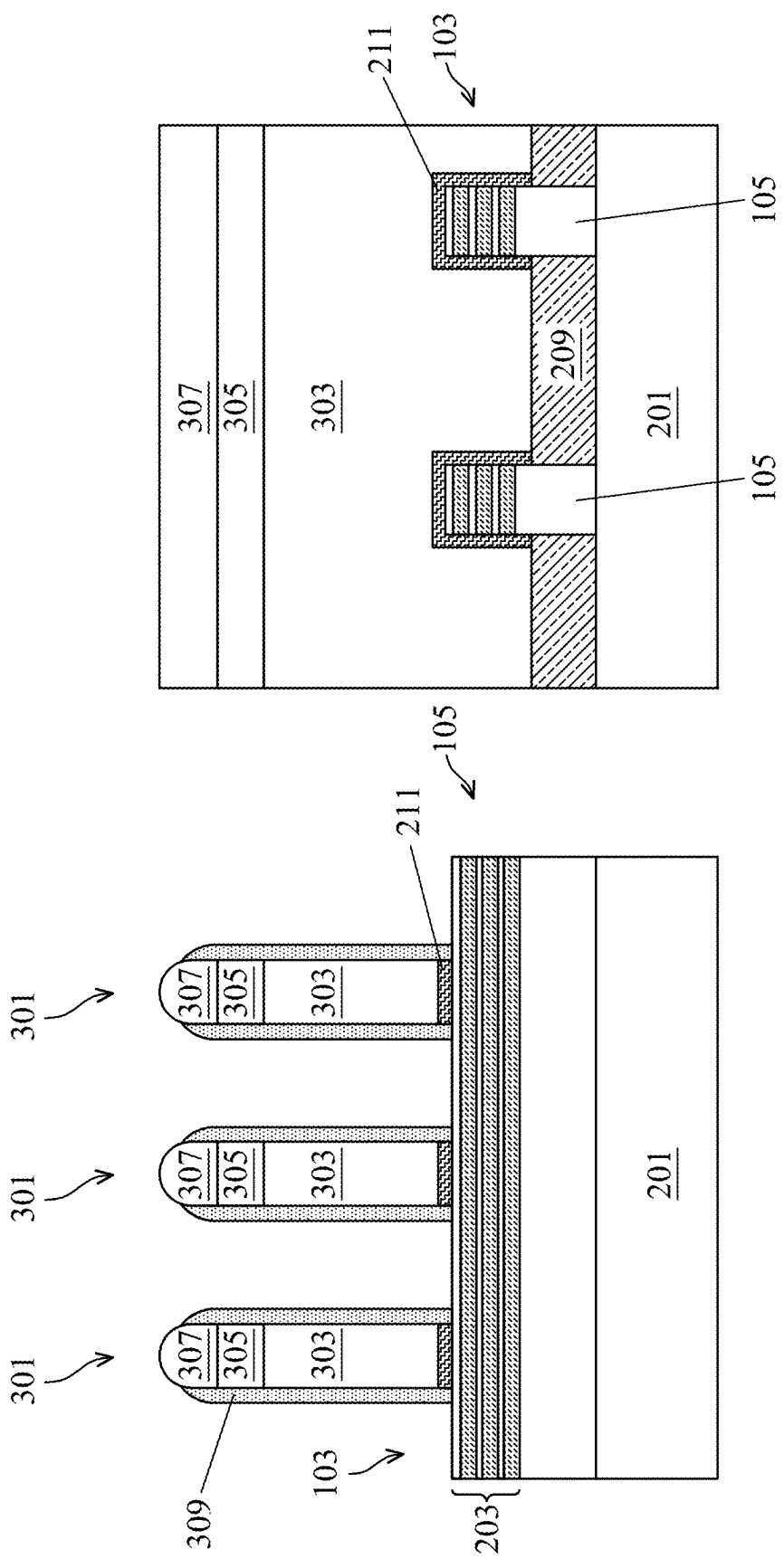
Y-cut

Figure 2B



X-cut

Figure 2A

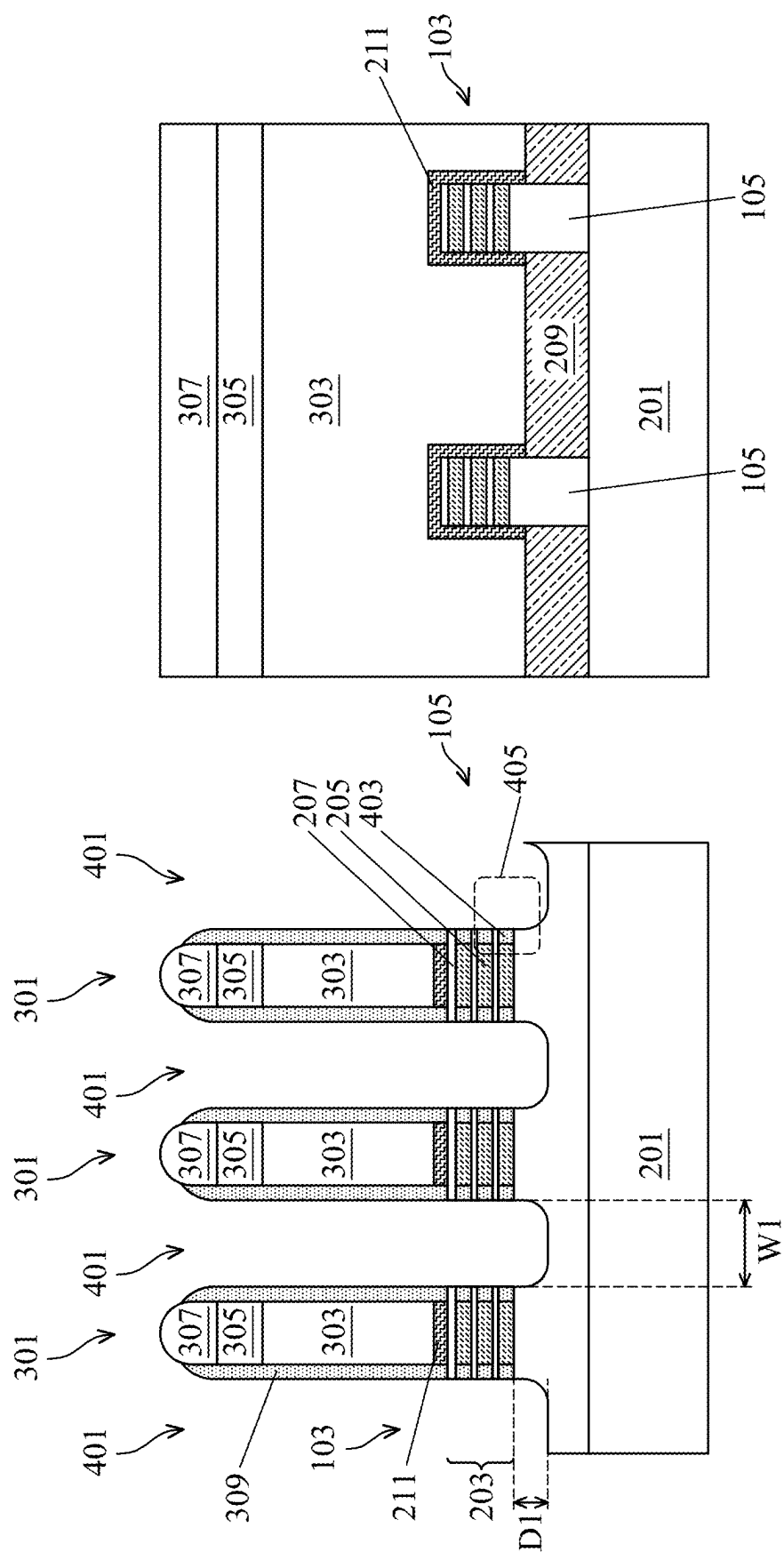


X-cut

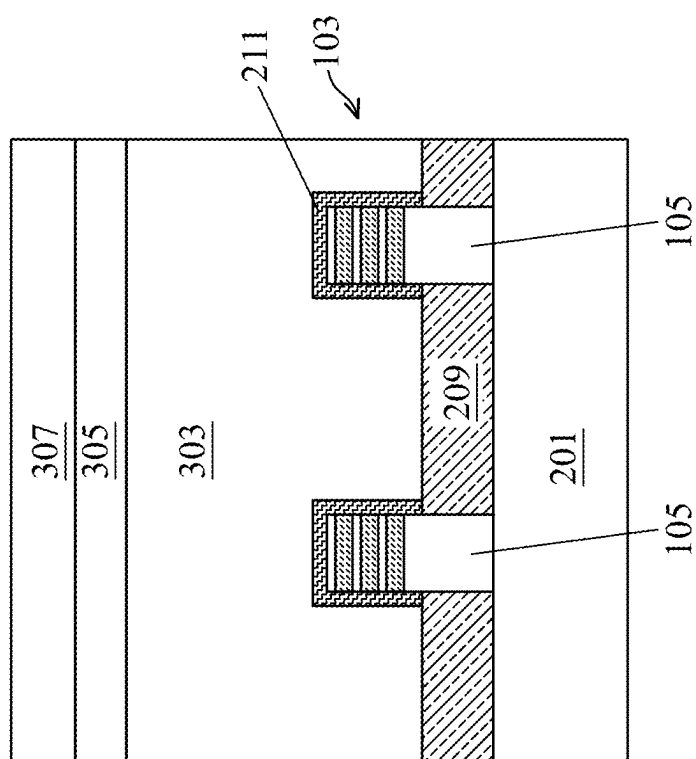
Figure 3A

Y-cut

Figure 3B



X-cut



Y-cut

Figure 4A

Figure 4B

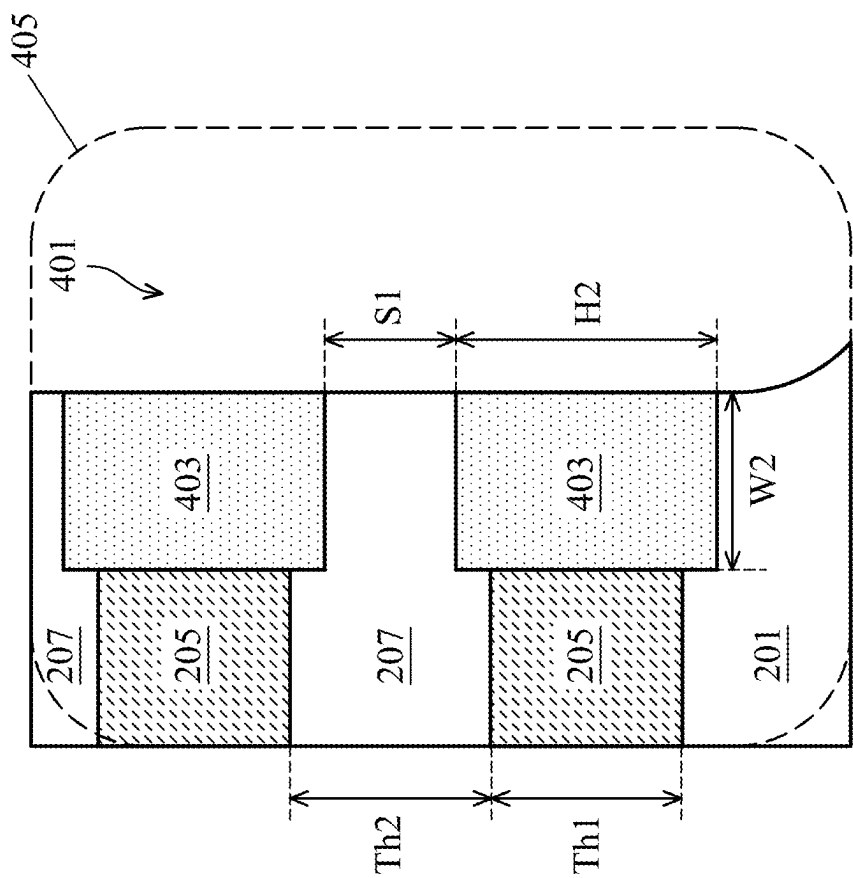
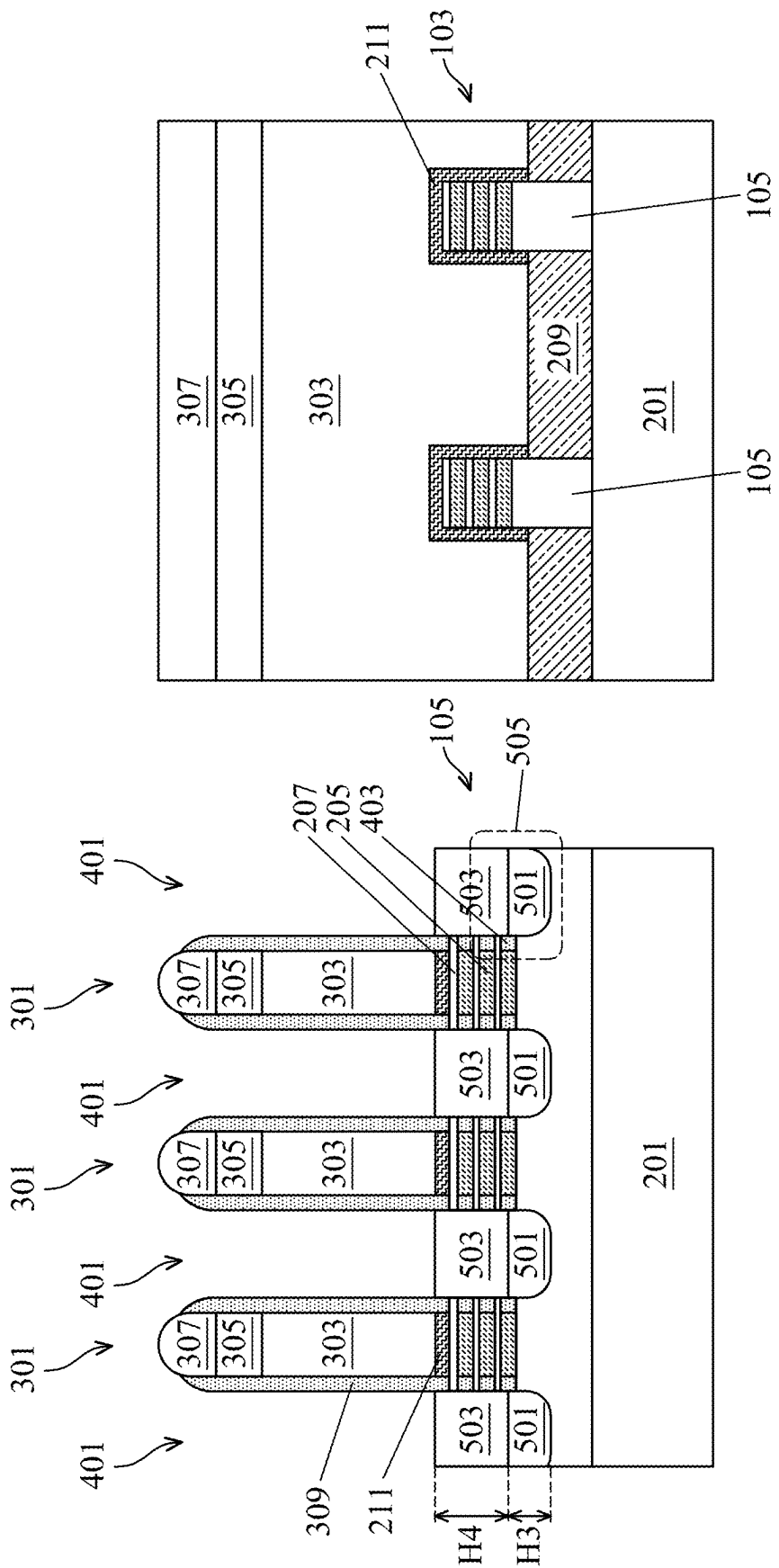
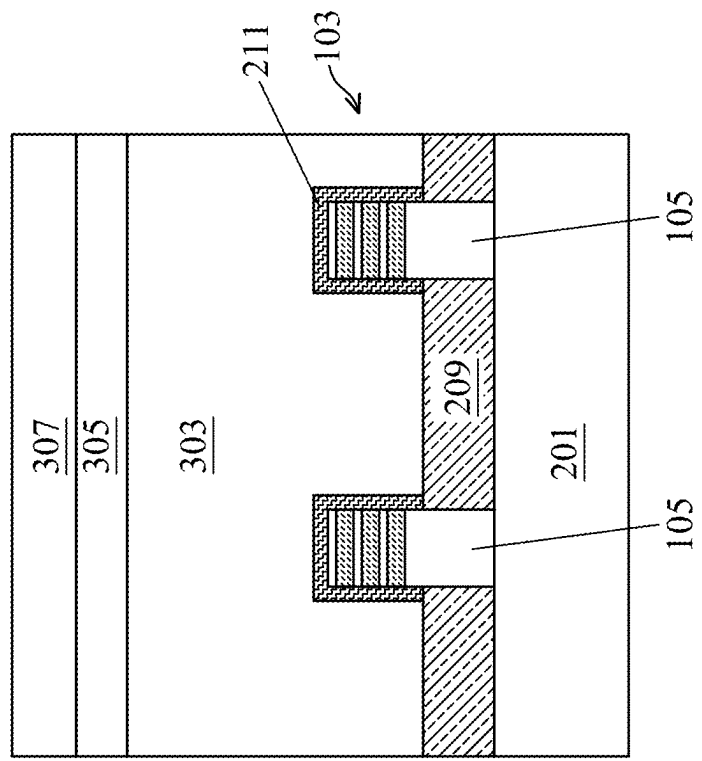


Figure 4C



X-cut

Figure 5A



Y-cut

Figure 5B

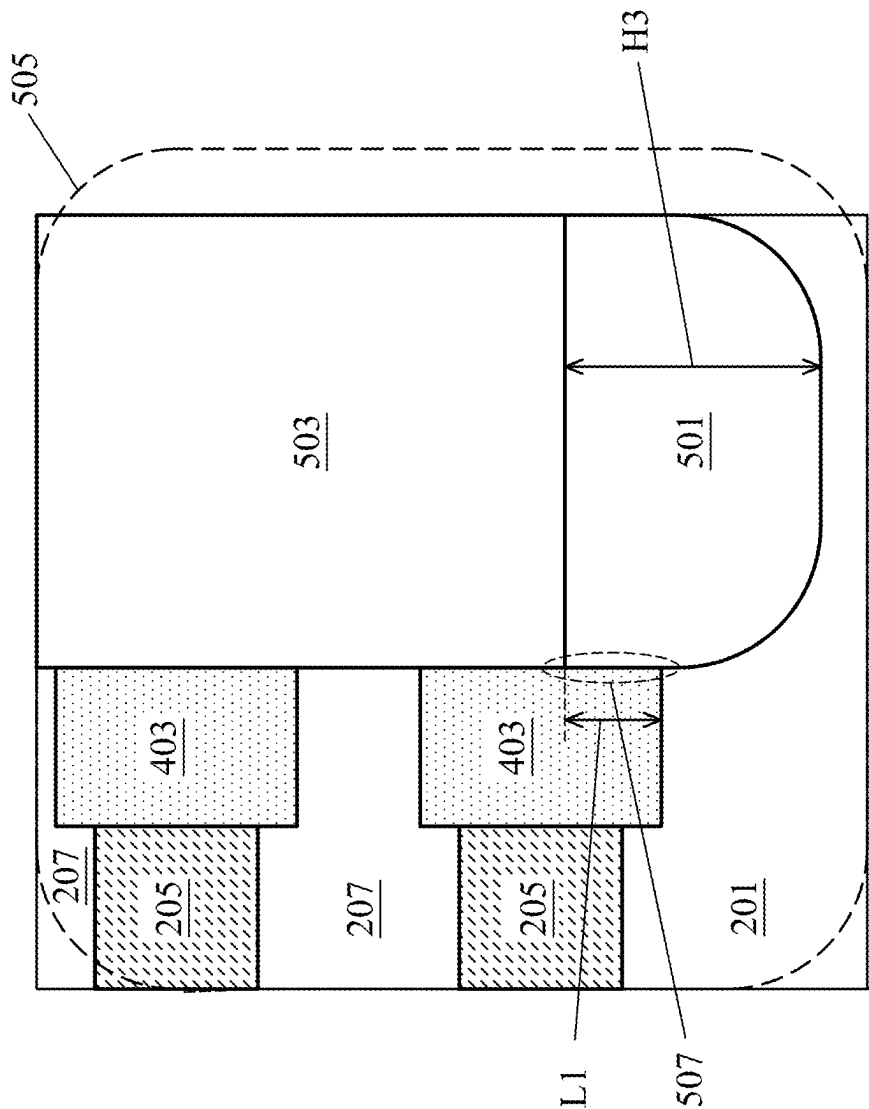


Figure 5C

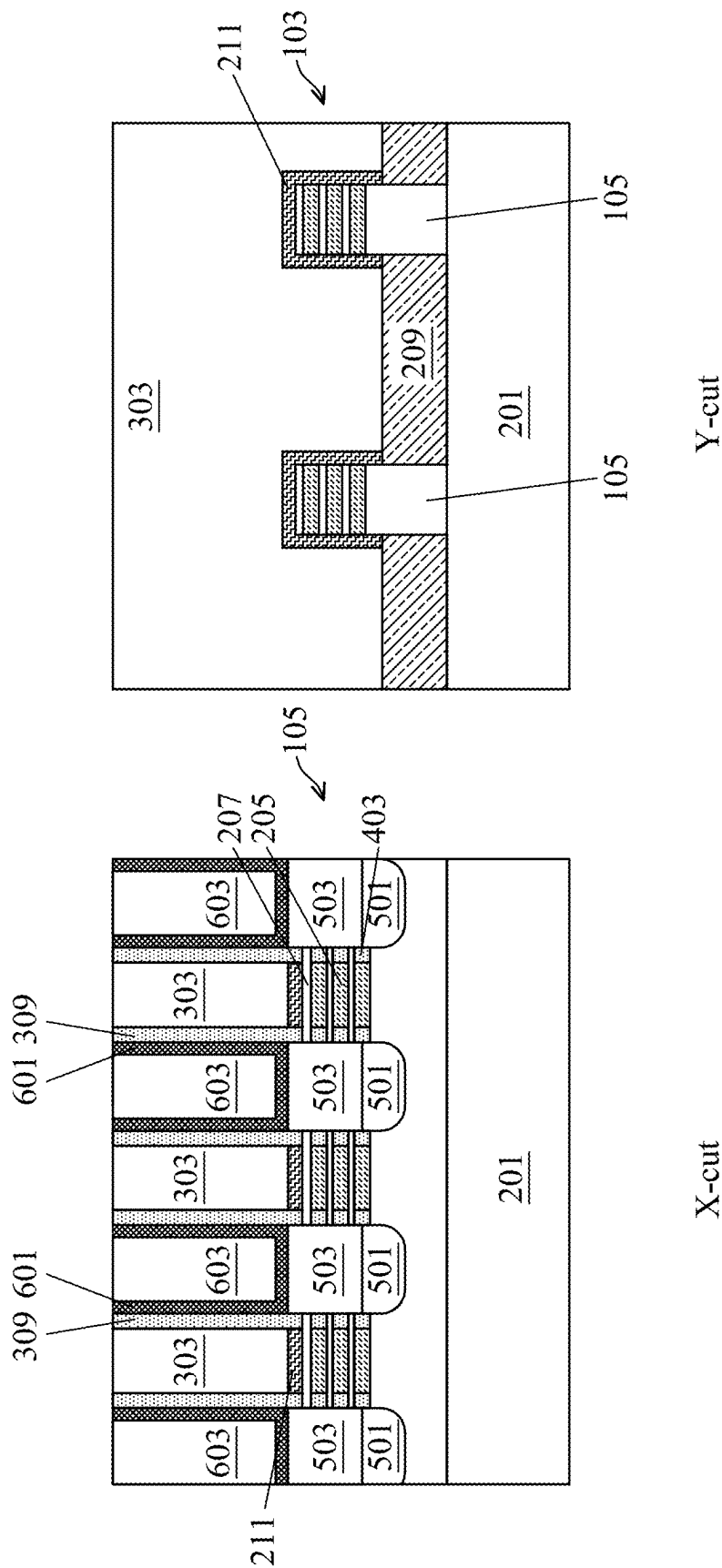


Figure 6A

Figure 6B

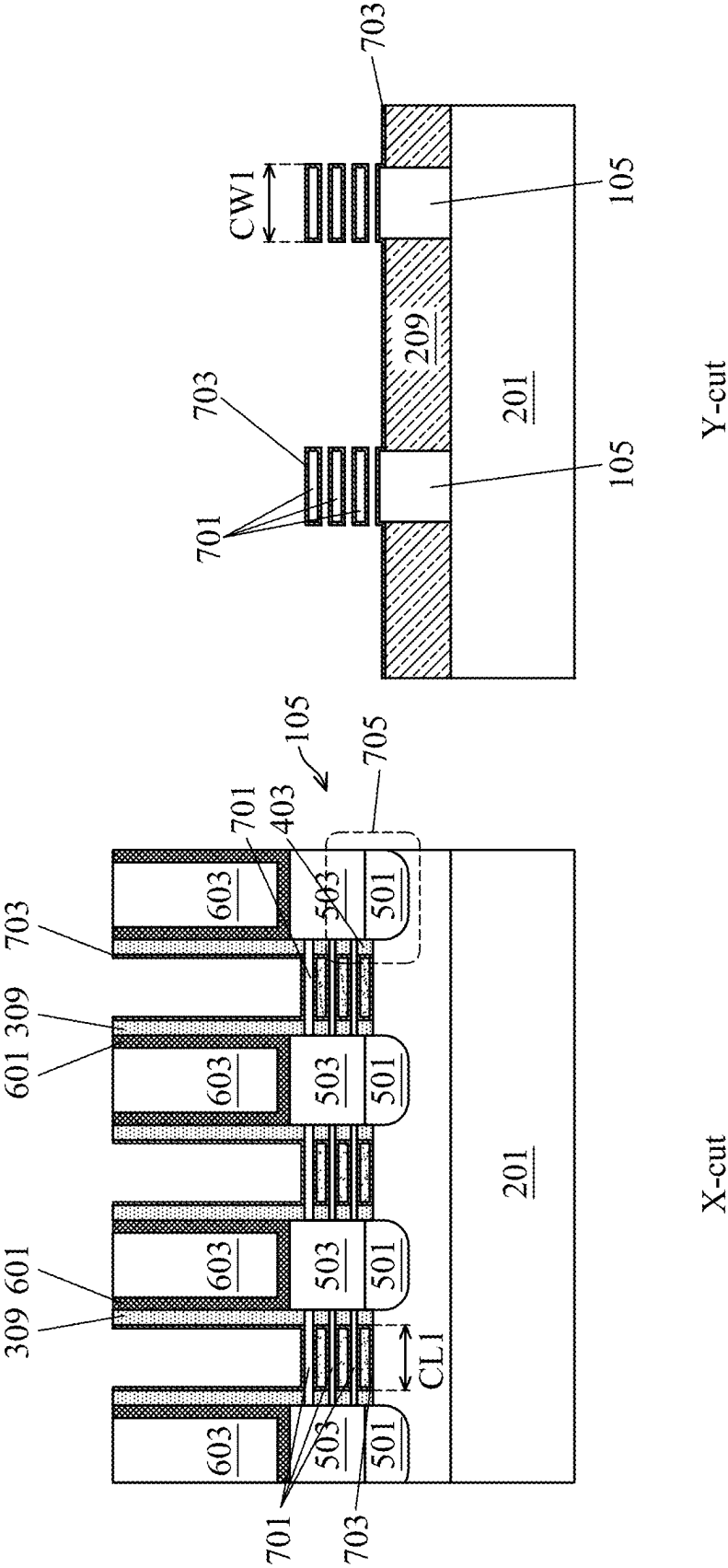


Figure 7A

Figure 7B

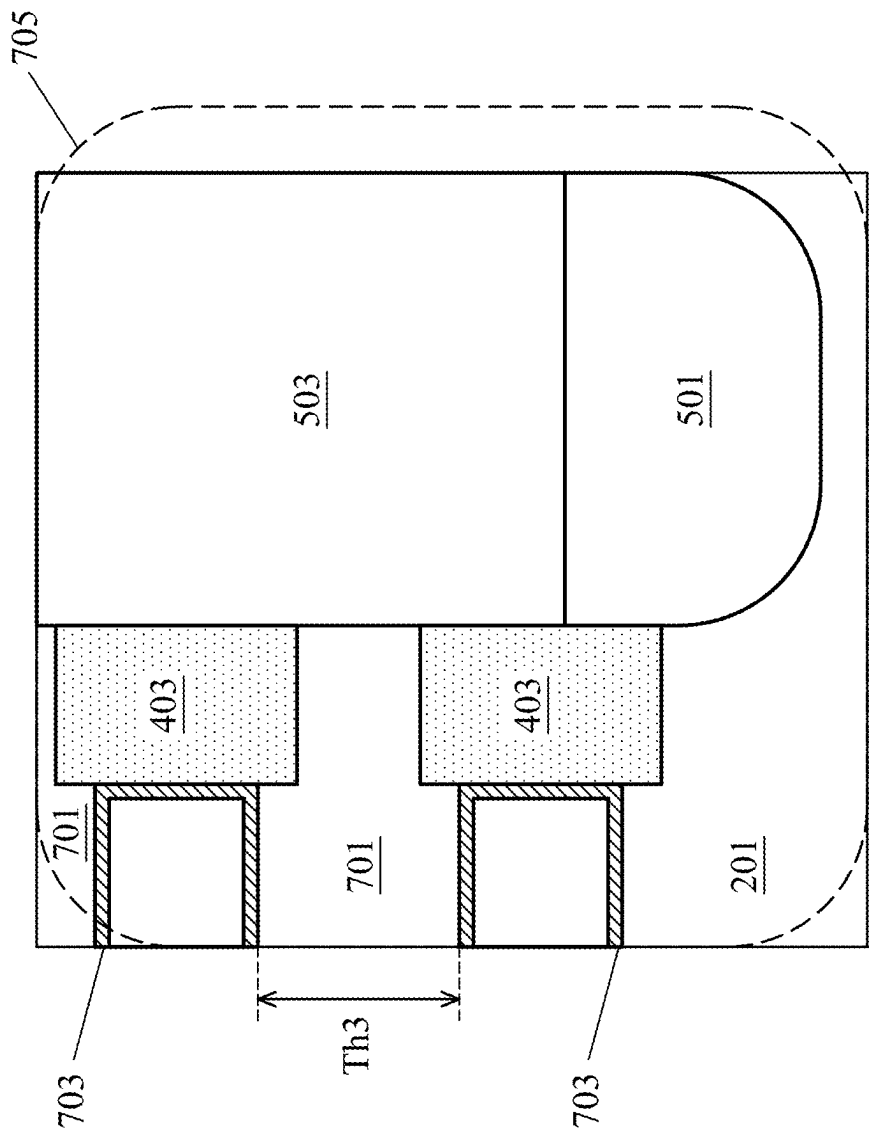


Figure 7C

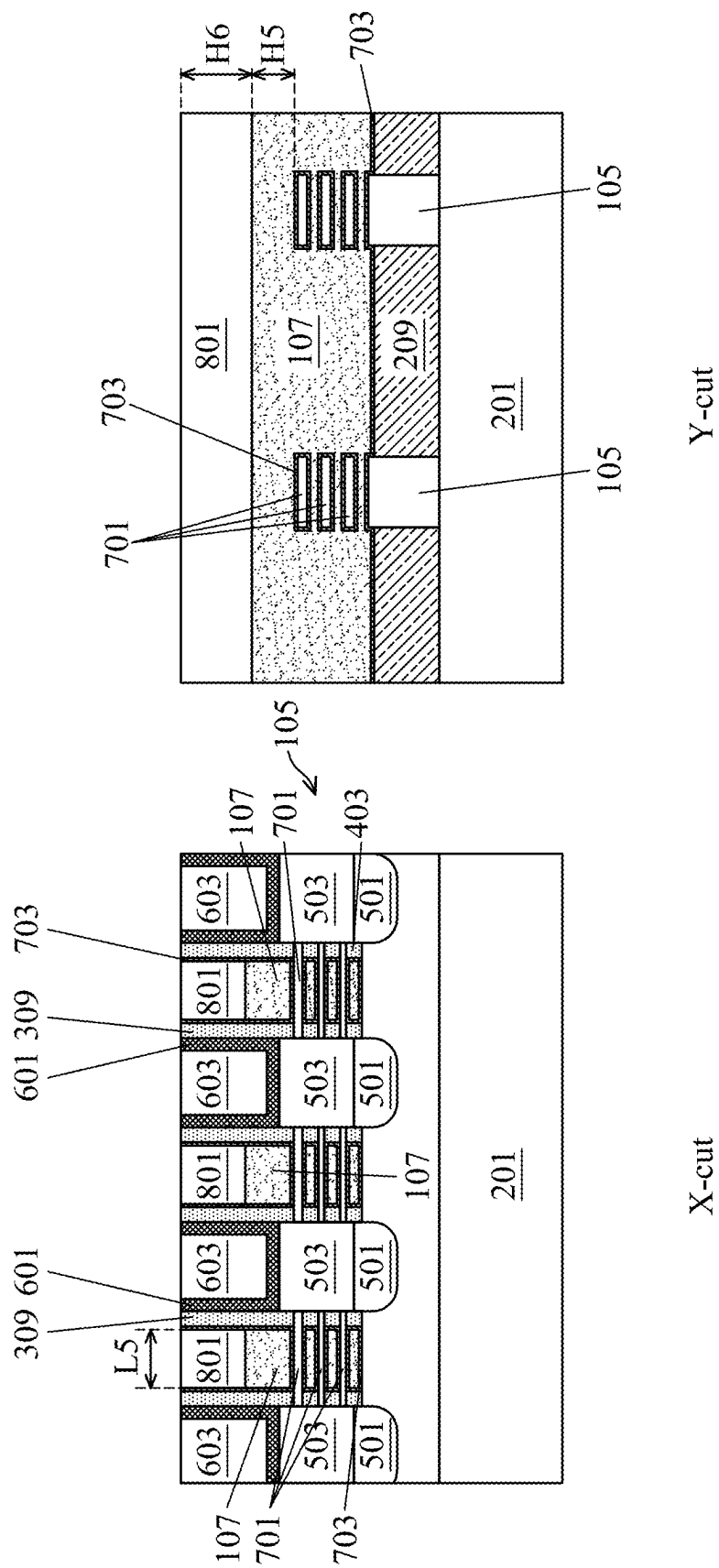


Figure 8A

Figure 8B

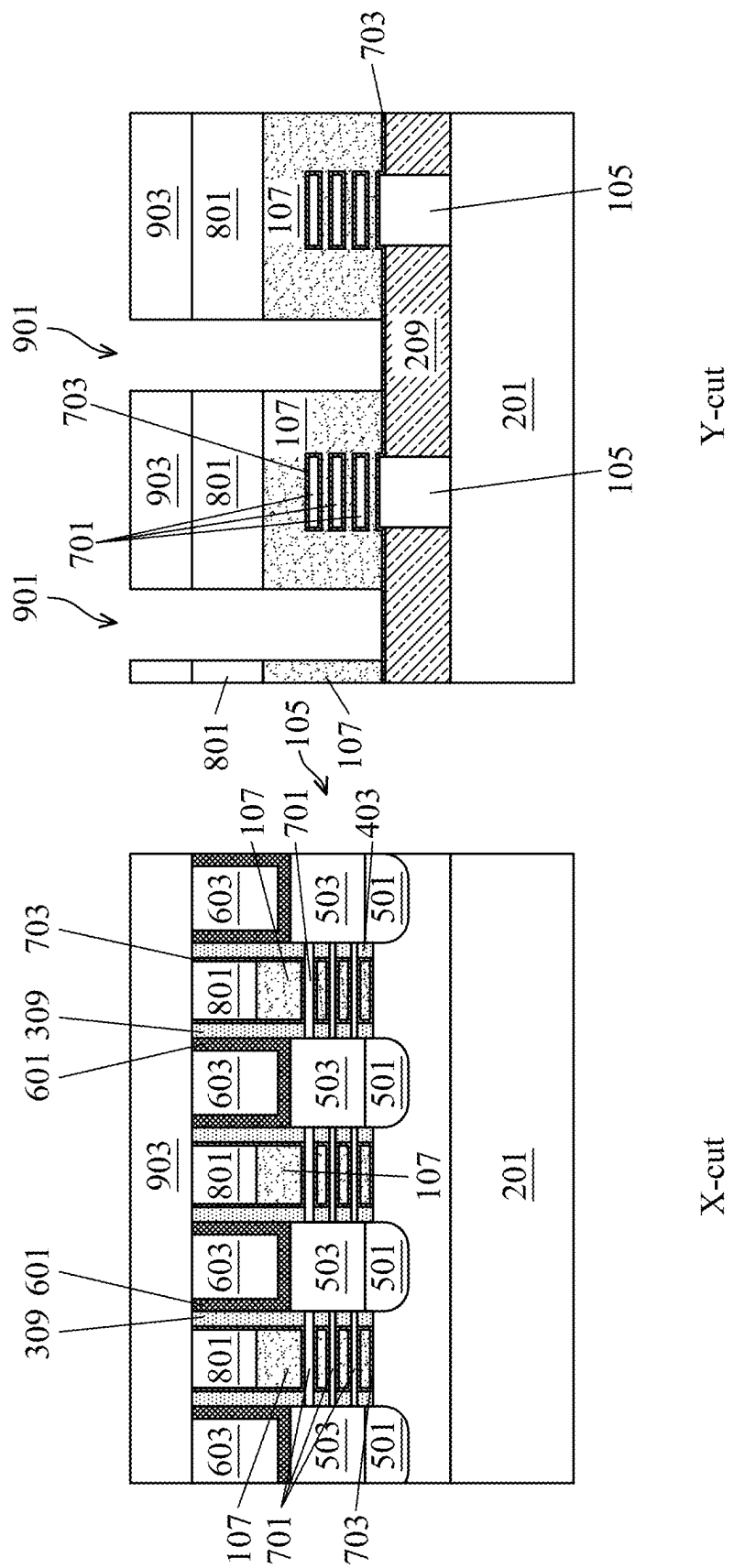


Figure 9A

Figure 9B

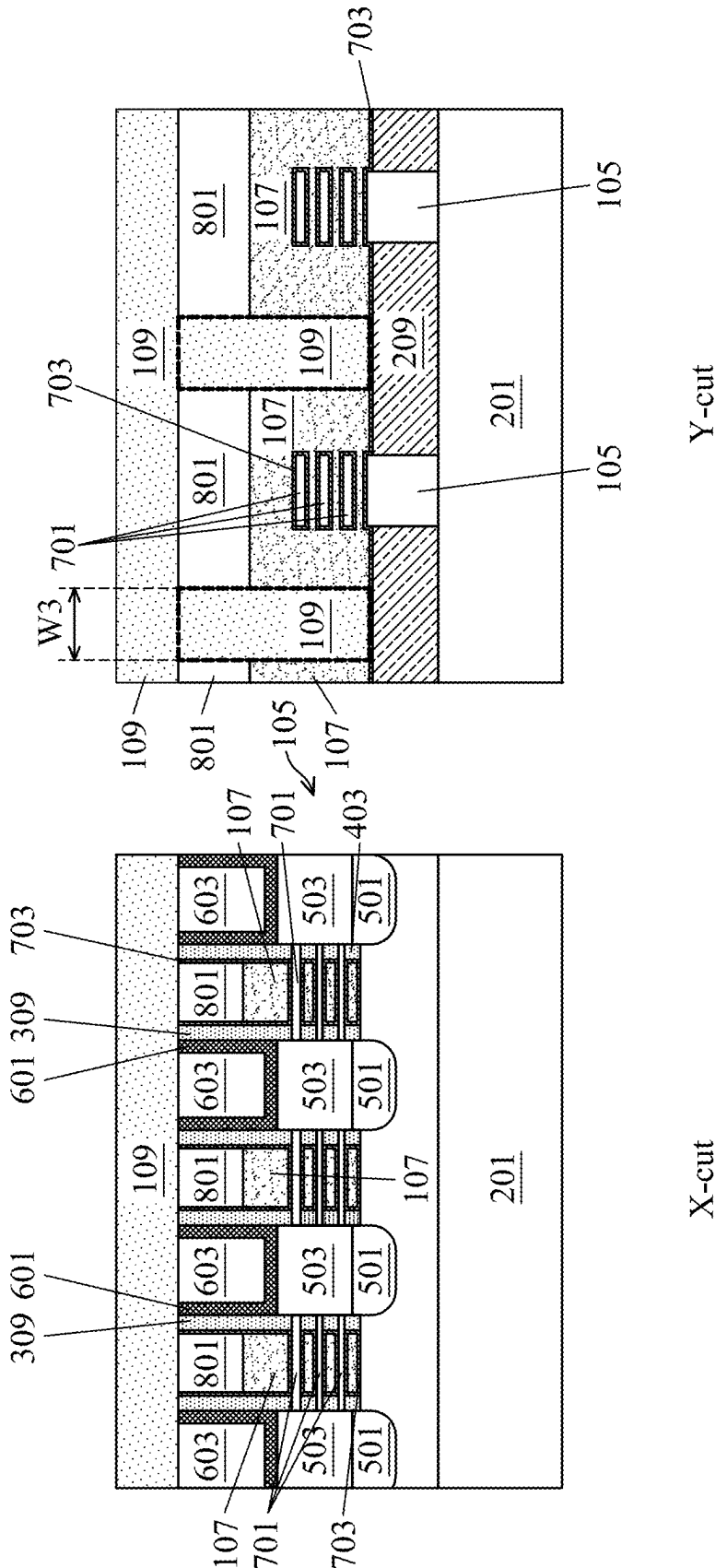


Figure 10A

Figure 10B

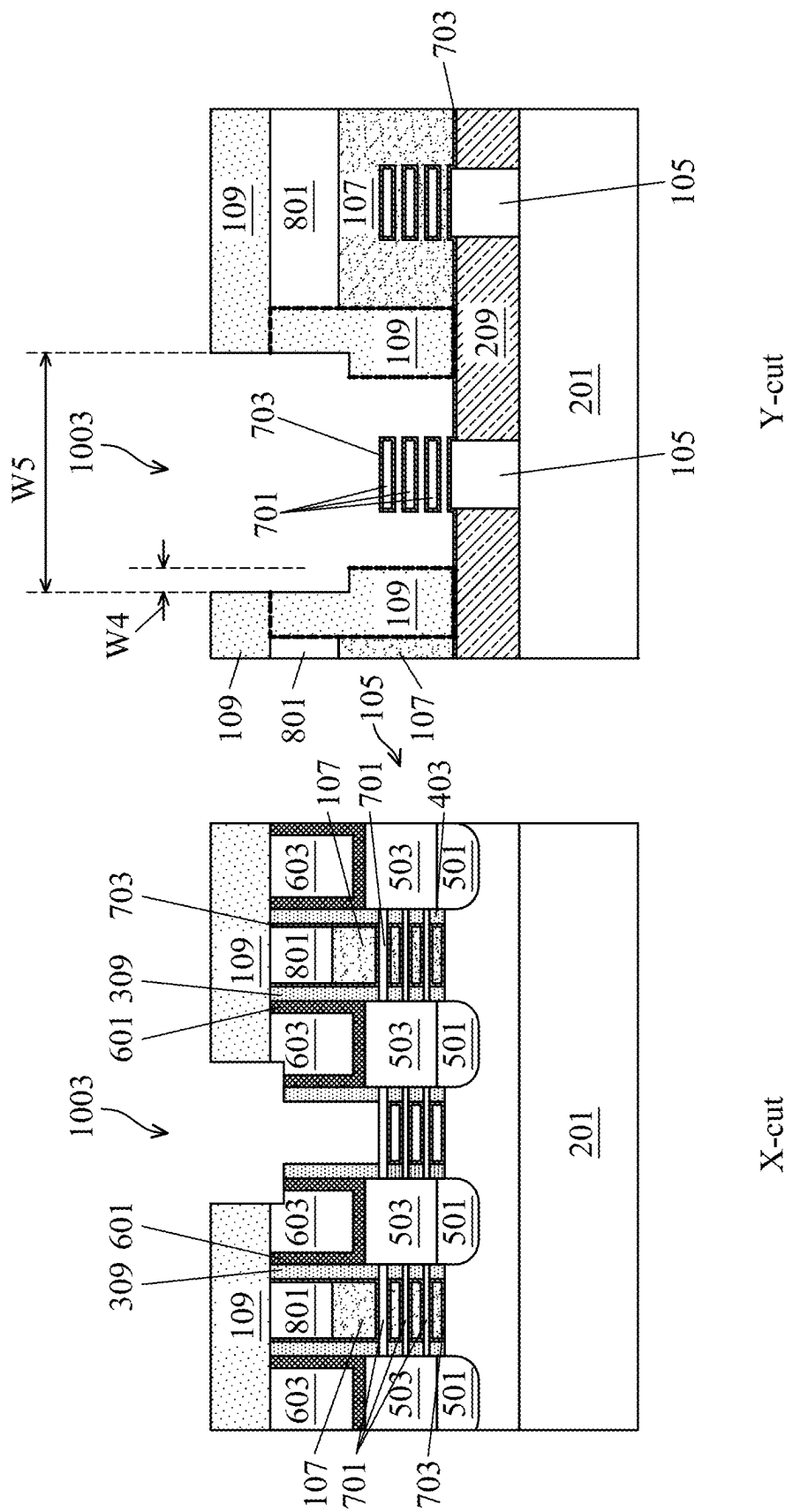


Figure 11A

Figure 11B

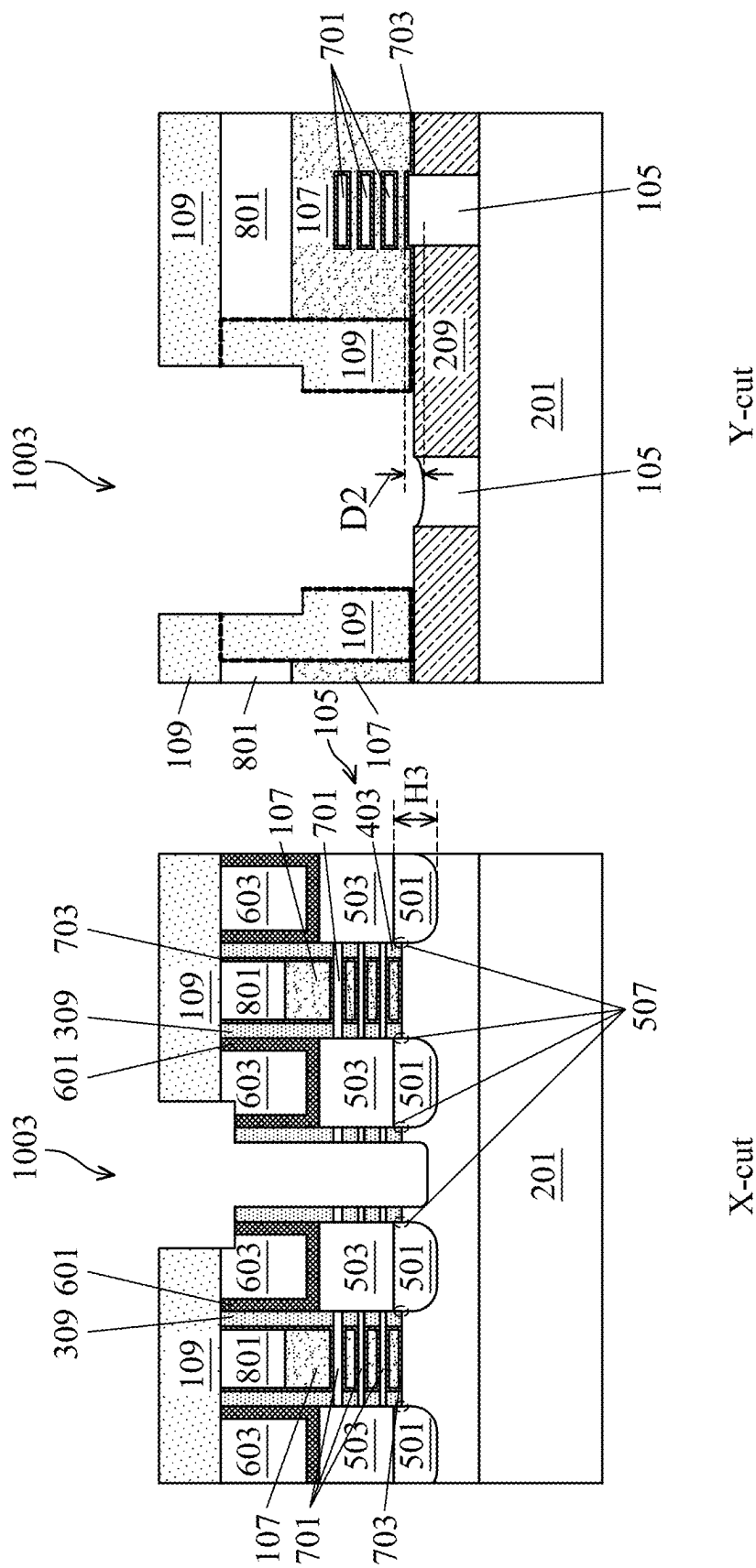
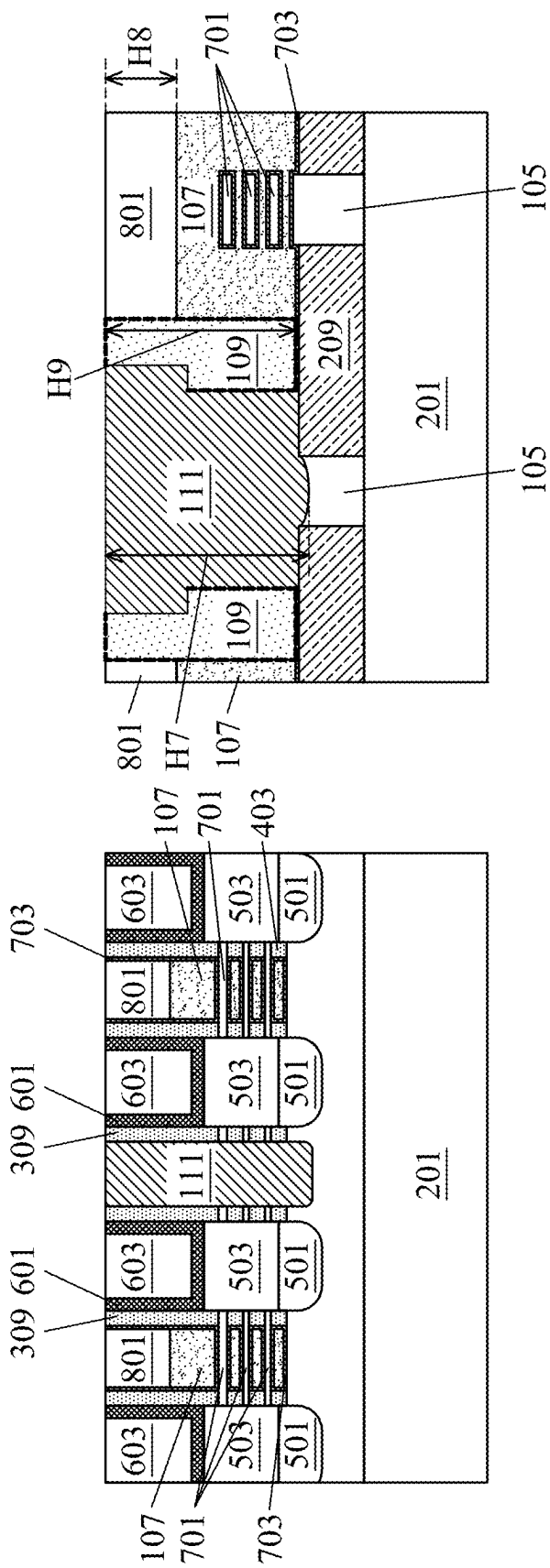


Figure 12A

Figure 12B

100 ↗



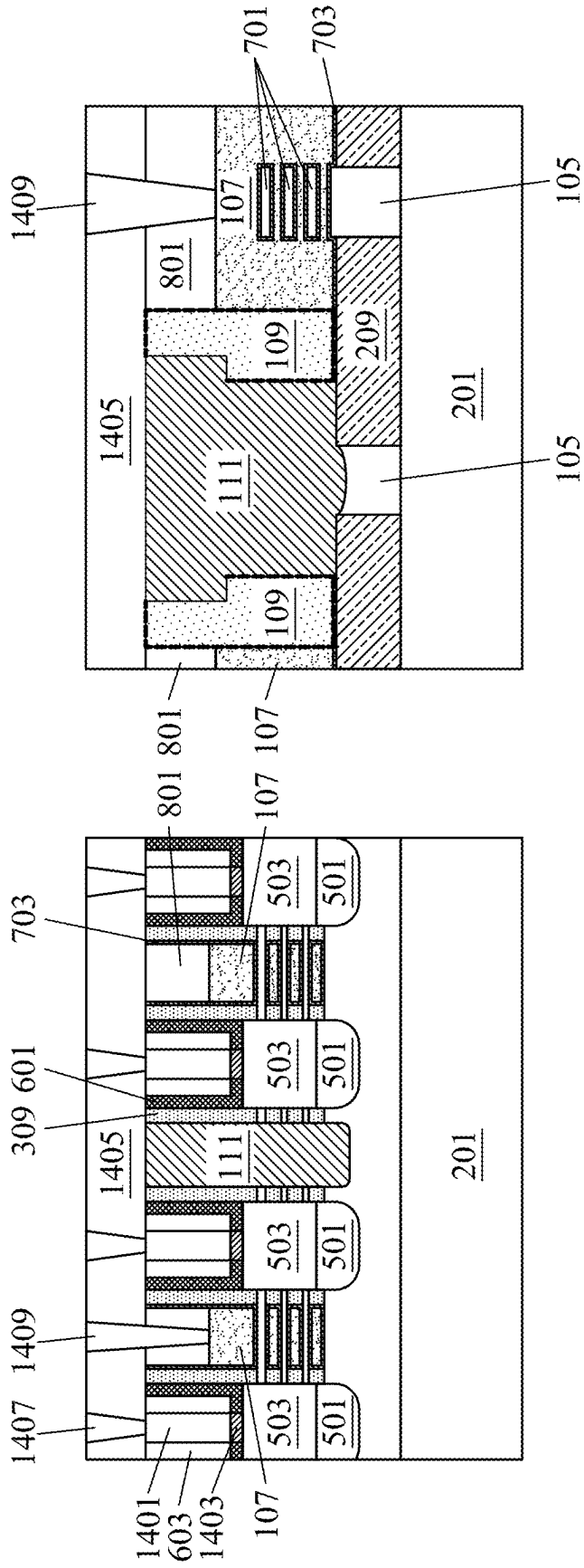
X-cut

Y-cut

Figure 13A

Figure 13B

1400 ↗



X-cut

Y-cut

Figure 14A

Figure 14B

SEMICONDUCTOR DEVICES AND METHODS OF MANUFACTURE

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 17/881,866, entitled: “Semiconductor Devices and Methods of Manufacture,” filed on Aug. 5, 2022, which is a divisional of U.S. patent application Ser. No. 16/872,058, entitled: “Semiconductor Devices and Methods of Manufacture,” filed on May 11, 2020, now U.S. Pat. No. 11,450,754 issued Sep. 20, 2022, which claims the benefit of U.S. Provisional Application No. 62/927,582, entitled: “Shallow CPODE Etching for GAA with SSD BT Spa,” filed on Oct. 29, 2019, which applications are hereby incorporated herein by reference.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

[0003] The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates a top-down view of a gate-all-around (GAA) semiconductor device, according to some embodiments.

[0006] FIGS. 2A-2B illustrates cross-sectional views of forming a multi-layer structure and forming fins in the multi-layer structure, in accordance with some embodiments.

[0007] FIGS. 3A-3B illustrates cross-sectional views of forming dummy gate stacks over the fins, in accordance with some embodiments.

[0008] FIGS. 4A-4C illustrates cross-sectional views of forming openings and inner spacers in the multi-layer structure, in accordance with some embodiments.

[0009] FIGS. 5A-5C illustrates cross-sectional views of forming bottom spacers and source/drain regions, in accordance with some embodiments.

[0010] FIGS. 6A-6B illustrates cross-sectional views of forming an inter-layer dielectric (ILD) layer and planarization process, in accordance with some embodiments.

[0011] FIGS. 7A-7C illustrates cross-sectional views of removing the dummy gate stacks, performing a wire-release process, and forming a gate dielectric in accordance with some embodiments.

[0012] FIGS. 8A-8B illustrates cross-sectional views of forming gate electrodes and gate caps, in accordance with some embodiments.

[0013] FIGS. 9A-9B illustrates cross-sectional views of forming openings in a cut-metal gate process, in accordance with some embodiments.

[0014] FIGS. 10A-10B illustrates cross-sectional views of forming cut-metal gate dielectric plugs, in accordance with some embodiments.

[0015] FIGS. 11A-11B illustrates cross-sectional views of forming an opening in an intermediate step in forming a Continuous Poly On Diffusion Edge (CPODE) structure, in accordance with some embodiments.

[0016] FIGS. 12A-12B illustrates cross-sectional views of extending the openings in a further intermediate step in forming the CPODE structure, in accordance with some embodiments.

[0017] FIGS. 13A-13B illustrates cross-sectional views of forming the CPODE structure in the GAA semiconductor device, in accordance with some embodiments.

[0018] FIGS. 14A-14B illustrates cross-sectional views of forming source/drain contact plugs, in accordance with some embodiments.

DETAILED DESCRIPTION

[0019] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0020] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0021] FIG. 1 illustrates a top-down view of an intermediate structure in forming a gate-all-around (GAA) semiconductor device 100, according to some embodiments. In particular, FIG. 1 illustrates a multi-layer structure 103 comprising a plurality of nanosheets formed over a semiconductor substrate 101 (illustrated in the following figures), fins 105 formed in the multi-layer structure 103, and a

plurality of gate electrodes **107** over the fins **105**. FIG. **1** further illustrates a plurality of cut-metal gate structures **109** separating two of the gate electrodes **107** and a Continuous Poly On Diffusion Edge (CPODE) structure **111** dividing one of the fins **105** in two and intersecting the cut-metal gate structures **109**.

[0022] Although two fins **105** are illustrated in FIG. **1** and in the following figures, it is understood that depending on the desired design and number of the GAA semiconductor devices **100**, any suitable number of fins **105** may be formed in the multi-layer structure **103** to form the desired GAA semiconductor devices **100**. Furthermore, any suitable number of gate electrodes **107**, CPODE structures **111**, and cut-metal gate structures **109** may be formed to form the desired GAA semiconductor devices **100**.

[0023] FIG. **1** further illustrates a first cutline (X-cut) and a second cutline (Y-cut) taken through the intermediate structure. The first cutline (X-cut) is taken through the length of the fin **105** divided in two and through the CPODE structure **111**. The second cutline (Y-cut) is taken through the length of one of the gate electrodes **107** separated by the two cut-metal gate structures **109**, through the two cut-metal gate structures **109**, and through the CPODE structure **111** intersecting the two cut-metal gate structures **109**.

[0024] FIGS. **2A-2B** illustrates cross-sectional views of forming the multi-layer structure **103** and forming the fins **105** in the multi-layer structure **103**, in an intermediate step of forming the GAA semiconductor device **100**, in accordance with some embodiments. In particular, FIG. **1** illustrates a substrate **201** into which dopants may be implanted in order to form wells. In an embodiment the substrate **201** is a semiconductor substrate, which may be, for example, a silicon substrate, a silicon germanium substrate, a germanium substrate, a III-V material substrate (e.g., GaAs, GaP, GaAsP, AlInAs, AlGaAs, GaInAs, InAs, GaInP, InP, InSb, and/or GaInAsP; or a combination thereof), or a substrate formed of other semiconductor materials with, for example, high band-to-band tunneling (BTBT). The substrate **201** may be doped or un-doped. In some embodiments, the substrate **201** may be a bulk semiconductor substrate, such as a bulk silicon substrate that is a wafer, a semiconductor-on-insulator (SOI) substrate, a multi-layered or gradient substrate, or the like.

[0025] FIGS. **2A-2B** further illustrate a deposition process to form the multi-layer structure **103** in an intermediate stage of manufacturing the GAA semiconductor device **100**, according to some embodiments. In particular, FIG. **2A** further illustrates a series of depositions that are performed to form a multi-layer stack **203** of alternating materials of first layers **205** and second layers **207** over the substrate **201**.

[0026] According to some embodiments, the first layers **205** may be formed using a first semiconductor material with a first lattice constant, such as SiGe, Ge, Si, GaAs, InSb, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, combinations of these, or the like. In some embodiments, a first layer **205** of the first semiconductor material (e.g., SiGe) is epitaxially grown on the substrate **201** using a deposition technique such as epitaxial growth, vapor-phase epitaxy (VPE), molecular beam epitaxy (MBE), although other deposition processes, such as chemical vapor deposition (CVD), low pressure CVD (LPCVD), atomic layer CVD (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), a combination thereof, or the like, may also be utilized. In some embodiments, the first layer **205** is formed

to first thicknesses Th1 of between about 3 nm and about 10 nm. However, any suitable thickness may be utilized while remaining within the scope of the embodiments.

[0027] Once the first layer **205** has been formed over the substrate **201**, a second layer **207** may be formed over the first layer **205**. According to some embodiments, the second layers **207** may be formed using a second semiconductor material such as Si, SiGe, Ge, GaAs, InSb, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, combinations of these, or the like with a second lattice constant that is different from the first lattice constant of the first layer **205**. In a particular embodiment in which the first layer **205** is silicon germanium, the second layer **207** is a material such as silicon. However, any suitable combination of materials may be utilized for the first layers **205** and the second layers **207**.

[0028] In some embodiments, the second layer **207** is epitaxially grown on the first layer **205** using a deposition technique similar to that used to form the first layer **205**. However, the second layer **207** may use any of the deposition techniques suitable for forming the first layer **205**, as set forth above or any other suitable technique. According to some embodiments, the second layer **207** is formed to a similar thickness to that of the first layer **205**. However, the second layer **207** may also be formed to a thickness that is different from the first layer **205**. According to some embodiments, the second layer **207** may be formed to a second thickness Th2 of between about 5 nm and about 15 nm. However, any suitable thickness may be used.

[0029] Once the second layer **207** has been formed over the first layer **205**, the deposition process is repeated to form the remaining material layers in the series of alternating materials of the first layers **205** and the second layers **207** until a desired topmost layer of the multi-layer stack **203** has been formed. According to the present embodiment, the first layers **205** may be formed to a same or similar first thickness and the second layers **207** may be formed to the same or similar second thickness. However, the first layers **205** may have different thicknesses from one another and/or the second layers **207** may have different thicknesses from one another and any combination of thicknesses may be used for the first layers **205** and the second layers **207**. According to the present embodiment, the topmost layer of the multi-layer stack **203** is formed as a second layer **207**; however, in other embodiments, the topmost layer of the multi-layer stack **203** may be formed as a first layer **205**. Additionally, although embodiments are disclosed herein comprising three of the first layers **205** and three of the second layers **207**, the multi-layer stack **203** may have any suitable number of layers (e.g., nanosheets). For example, the multi-layer stack **203** may comprise multiple nanosheets in a range between 2 to 10 nanosheets. In some embodiments, the multi-layer stack **203** may comprise equal numbers of the first layers **205** to the second layers **207**; however, in other embodiments, the number of the first layers **205** may be different from the number of the second layers **207**. According to some embodiments, the multi-layer stack **203** may be formed to a first height H1 of between about 12 nm and about 100 nm. However, any suitable height may be used.

[0030] FIGS. **2A-2B** further illustrate, a patterning process of the multi-layer structure **103** and a formation of isolation regions **209** in an intermediate stage of manufacturing the GAA semiconductor device **100**, in accordance with some embodiments. The patterning process is used to form fins **105** in the multi-layer structure **103** and to form trenches

between the fins **105** in preparation for forming the isolation regions **209**. The patterning process for forming the fins **105**, according to some embodiments, comprises applying a photoresist over the multi-layer stack **203** and then patterning and developing the photoresist to form a mask over the multi-layer stack **203**. Once formed, the mask is then used during an etching process, such as an anisotropic etching process to transfer the pattern of the mask into the underlying layers to form the trenches through the multi-layer stack **203** and into the substrate **201** to define the fins **105**, wherein the fins **105** are separated by the trenches.

[0031] Additionally, while a single mask process has been described, this is intended to be illustrative and is not intended to be limiting, as the gate all around (GAA) device structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0032] In an embodiment the isolation regions **209** are formed as shallow trench isolation regions by depositing a dielectric material in the trenches. According to some embodiments, the dielectric material used to form the isolation regions **209** may be a material such as an oxide material (e.g., a flowable oxide), high-density plasma (HDP) oxide, or the like. The dielectric material may be formed, after an optional cleaning and lining of the trenches, using either a chemical vapor deposition (CVD) method (e.g., the HARP process), a high density plasma CVD method, or other suitable method of formation to fill or overfill the regions around the fins **105**. In some embodiments, a post placement anneal process (e.g., oxide densification) is performed to densify the material of the isolation regions **209** and to reduce its wet etch rate. A chemical mechanical polishing (CMP), an etch, a combination of these, or the like may be performed to remove any excess material of the isolation regions **209**.

[0033] Once the dielectric material has been deposited to fill or overfill the regions around the fins **105**, the dielectric material may then be recessed away from the surface of the fins **105** to form the isolation regions **209**. The recessing may be performed to expose at least a portion of the sidewalls of the fins **105** adjacent to the top surface of the fins **105**. The dielectric material may be recessed using a wet etch by dipping the top surface of the fins **105** into an etchant selective to the material of the dielectric material, although other methods, such as a reactive ion etch, a dry etch, chemical oxide removal, or dry chemical clean may be used.

[0034] FIGS. 2A-2B further illustrate the formation of a dummy gate dielectric **211** over the exposed portions of the fins **105**. Once the isolation regions **209** have been formed, the dummy gate dielectric **211** may be formed by thermal oxidation, chemical vapor deposition, sputtering, or any other methods known and used in the art for forming a gate dielectric. Depending on the technique of gate dielectric

formation, the dummy gate dielectric **211** thickness on the top may be different from the dummy dielectric thickness on the sidewall. In some embodiments, the dummy gate dielectric **211** may be formed by depositing a material such as silicon and then oxidizing or nitridizing the silicon layer in order to form a dielectric such as the silicon dioxide or silicon oxynitride. In such embodiments, the dummy gate dielectric **211** may be formed to a thickness ranging from between about 3 Å and about 100 Å, such as about 10 Å. In other embodiments, the dummy gate dielectric **211** may also be formed from a high permittivity (high-k) material such as lanthanum oxide (La₂O₃), aluminum oxide (Al₂O₃), hafnium oxide (HfO₂), hafnium oxynitride (HfON), or zirconium oxide (ZrO₂), or combinations thereof, with an equivalent oxide thickness of between about 0.5 Å and about 100 Å, such as about 10 Å or less. Additionally, any combination of silicon dioxide, silicon oxynitride, and/or high-k materials may also be used for the dummy gate dielectric **211**.

[0035] Turning to FIGS. 3A-3B, these figures illustrate the formation of dummy gate stacks **301** over the fins **105**, in accordance with some embodiments. According to some embodiments, the dummy gate stacks **301** comprise the dummy gate dielectric **211**, a dummy gate electrode **303** over the dummy gate dielectric **211**, a first hard mask **305** over the dummy gate electrode **303**, and a second hard mask **307** over the first hard mask **305**.

[0036] In some embodiments, the dummy gate electrode **303** comprises a conductive material and may be selected from a group comprising of polysilicon, W, Al, Cu, AlCu, W, Ti, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ta, TaN, Co, Ni, combinations of these, or the like. The dummy gate electrode **303** may be deposited by chemical vapor deposition (CVD), sputter deposition, or other techniques known and used in the art for depositing conductive materials. The thickness of the dummy gate electrode **303** may be in the range of about 5 Å to about 500 Å. The top surface of the dummy gate electrode **303** may have a non-planar top surface, and may be planarized prior to patterning of the dummy gate electrode **303** or gate etch. Ions may or may not be introduced into the dummy gate electrode **303** at this point. Ions may be introduced, for example, by ion implantation techniques.

[0037] Once the dummy gate electrode **303** has been formed, the dummy gate dielectric **211** and the dummy gate electrode **303** may be patterned. In an embodiment the patterning may be performed by initially forming a first hard mask **305** over the dummy gate electrode **303** and forming the second hard mask **307** over the first hard mask **305**.

[0038] According to some embodiments, the first hard mask **305** comprises a dielectric material such as silicon nitride (SiN), oxide (OX), silicon oxide (SiO), titanium nitride (TiN), silicon oxynitride (SiON), combinations of these, or the like. The first hard mask **305** may be formed using a process such as chemical vapor deposition, plasma enhanced chemical vapor deposition, atomic layer deposition, or the like. However, any other suitable material and method of formation may be utilized. The first hard mask **305** may be formed to a thickness of between about 20 Å and about 3000 Å, such as about 20 Å.

[0039] The second hard mask **307** comprises a separate dielectric material from the material of the first hard mask **305**. The second hard mask **307** may comprise any of the materials and use any of the processes suitable for forming the first hard mask **305** and may be formed to a same or

similar thickness as the first hard mask **305**. In embodiments where the first hard mask **305** comprises silicon nitride (SiN), the second hard mask **307** may be e.g., an oxide (OX). However, any suitable dielectric materials, processes and thicknesses may be used to form the second hard mask.

[0040] Once the first hard mask **305** and the second hard mask **307** have been formed, the first hard mask **305** and the second hard mask **307** may be patterned. In an embodiment the first hard mask **305** and the second hard mask **307** may be patterned by initially placing a photoresist (not individually illustrated) over the first hard mask **305** and the second hard mask **307** and exposing the photoresist to a patterned energy source (e.g., light) in order to initiate a chemical reaction that modifies the physical properties of the exposed portions of the first photoresist. The first photoresist may then be developed by applying a first developer (also not individually illustrated) in order to utilize the modified physical properties between the exposed region and the unexposed region to selectively remove either the exposed region or the unexposed region.

[0041] Once the photoresist has been patterned, the photoresist may be used as a mask in order to pattern the underlying first hard mask **305** and the second hard mask **307**. In an embodiment the first hard mask **305** and the second hard mask **307** may be patterned using, e.g., one or more reactive ion etching (RIE) processes with the photoresist as a mask. The patterning process may be continued until the dummy gate electrode **303** is exposed beneath the first hard mask **305**.

[0042] Once the first hard mask **305** and the second hard mask **307** have been patterned, the photoresist may be removed utilizing, e.g., an ashing process, whereby a temperature of the photoresist is raised until the photoresist experiences a thermal decomposition and may be easily removed using one or more cleaning process. However, any other suitable removal process may be utilized.

[0043] Once the first hard mask **305** and the second hard mask **307** have been patterned, the dummy gate electrode **303** and the dummy gate dielectric **211** may be patterned in order to form a series of dummy gate stacks **301**. In an embodiment the dummy gate electrode **303** and the dummy gate dielectric **211** are patterned using an anisotropic etching process, such as a reactive ion etch, although any suitable process may be utilized.

[0044] FIGS. 3A-3B further illustrate a formation of top spacers **309**. According to an embodiment, a spacer material is formed by blanket deposition on the dummy gate stacks **301** and the multi-layer structure **103**. As such, the spacer material is deposited over the top surfaces and sidewalls of the dummy gate stacks **301**, the top surfaces and sidewalls of the fins **105** and top surfaces of the isolation regions **209**. According to some embodiments, the spacer material comprises a dielectric material and is formed may be formed using methods such as chemical vapor deposition (CVD), plasma enhanced CVD, sputter, thermal oxidation, and any other suitable methods. According to some embodiments, the spacer material comprises materials such as silicon oxide (SiO₂), silicon oxynitride (SiON), silicon nitride (Si₃N₄), silicon oxycarbide (SiOC), silicon oxycarbonitride (SiOCN), although any suitable material, such as low-k materials with a k-value less than about 4.0, combination thereof, or the like may be utilized.

[0045] Once formed, the spacer material may be etched in order to shape the top spacers **309** on the dummy gate stacks

301 and the fins **105** to expose the tops of the dummy gate stacks **301** and fins **105**. According to some embodiments, the spacer material may be etched using an anisotropic etching process (e.g., a dry etching process) such as a reactive ion etching (RIE) process, an isotropic etching process (e.g., a wet etching process), combination thereof, or the like. In some embodiments, the spacer material formed over the fins **105** in source/drain regions may be recessed during the etching process and/or during a subsequent etching process such that portions along the sidewalls of the fins **105** in those source/drain regions are exposed.

[0046] However, while embodiments are described using a single spacer material, this is intended to be illustrative and is not intended to be limiting. Rather, any number of spacer materials and any combinations of deposition and removal processes may be used, and all such processes are fully intended to be included within the scope of the embodiments.

[0047] FIGS. 4A to 4C illustrate cross-sectional views of forming first openings **401** and inner spacers **403** in the fins **105**, in accordance with some embodiments. During the formation of the top spacers **309**, the multi-layer stack **203** will be re-exposed by the removal of the spacer material used for the top spacers **309**. Once exposed, an etching process to remove material from the multi-layer stack **203** and the substrate may be performed to form the first openings **401** (e.g., trenches, recesses, channels or the like) which extend through the multi-layer stack **203** and into the semiconductor substrate **101** in preparation for forming source/drain regions therein. In an embodiment, the etching may be performed using one or more anisotropic etches, such as reactive ion etches, although any suitable processes may be utilized.

[0048] In an embodiment, the first openings **401** may be formed to have a first width W1 of between about 10 nm and about 40 nm, such as about 20 nm. Additionally, the first openings **401** may be formed to extend into the substrate **101** a first depth D1 of between about 3 nm and about 40 nm, such as about 10 nm. However, any suitable dimensions may be utilized.

[0049] FIGS. 4A-4C further illustrate the formation of inner spacers **403** in the first layers **205**. In particular, FIG. 4C illustrates a magnified view a section **405** that is highlighted in FIG. 4A by a dashed line and may be referenced with the following discussion with regard to the formation of the inner spacers **403**. In some embodiments, the inner spacers **403** are formed by patterning recesses using a wet etch with an etchant that is more selective to the material of the first layers **205** (e.g., silicon germanium (SiGe)) than the material of the second layers **207** (e.g., silicon (Si)) or the substrate **201** (e.g., silicon (Si)). For example, in an embodiment in which the first layers **205** are silicon germanium and the second layers **207** are silicon, the wet etch may use an etchant such as hydrochloric acid (HCl).

[0050] In an embodiment the wet etching process may be a dip process, a spray process, a spin-on process, or the like and may be performed using any suitable process temperatures (e.g., between about 400° C. and about 600° C.) and any suitable process times (e.g., between about 100 seconds and about 1000 seconds, such as about 300 seconds). However, any suitable process conditions and parameters may be utilized. The etching process may be continued such that recesses with facet limited surfaces are formed in each

of the first layers **205** to any desired distance from the sidewall of the first openings **401**.

[0051] Additionally, the selectivity of the etchant may be chosen. According to some embodiments, the etching process may also be performed to remove portions of the second layers **207** and the substrate **201** such that a size of the recesses extends vertically in the direction of the sidewall of the first openings **401** to any suitable height (described further below).

[0052] However, a wet etching process is not the only process that may be utilized to recess the first layers **205**. For example, in another embodiment the recessing of the first layers **205** may be performed with an isotropic dry etching process or a combination of a dry etching process and a wet etching process. Any suitable process of recessing the first layers **205** may be utilized, and all such processes are fully intended to be included within the scope of the embodiments.

[0053] Once the recesses are formed in each of the first layers **205**, a spacer material is formed over the multi-layer structure **103**. In some embodiments, the spacer material can be different from the material of the top spacers **309** and can be a dielectric material comprising silicon such as silicon nitride (SiN), silicon oxide (SiO₂), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), although any suitable material such as low-k materials with a k-value less than about 4.0, or even an air gap, or combination thereof may also be utilized. The spacer material may be deposited using a deposition process such as chemical vapor deposition, physical vapor deposition, or atomic layer deposition to a thickness of between about 2 nm and about 10 nm, such as about 5 nm. However, any suitable thickness or deposition process may be utilized.

[0054] By depositing the spacer material over the **103**, the spacer material will line the sidewalls of the first openings **401** and will also fill in the recesses in the first layers **205**. Once the recesses have been filled with the spacer material, a removal process is then performed to remove any excess spacer material from the first openings **401**, while leaving behind the inner spacers **403**. In an embodiment, the removal of the excess spacer material may be performed using an etching process such as, e.g., an anisotropic, dry etching process such as a reactive ion etching process. However, any suitable etching process, which removes the excess spacer material from the first openings **401** while leaving behind the inner spacers **403**, may be utilized.

[0055] By filling the recesses with the spacer material and removing the excess spacer material from the first openings **401**, the inner spacers **403** will take on the shape of the recesses. Additionally, while an embodiment forming the inner spacers **403** to faceted shapes is described, this is intended to be illustrative and is not intended to be limited. Rather, any suitable shape, such as a concave shape or a convex shape, or even the inner spacers **403** being recessed may be utilized. All such shapes are fully intended to be included within the scope of the embodiments. According to some embodiments, the inner spacers **403** may be formed to a second width W2 of between about 2 nm and about 10 nm, such as about 5 nm and a second height H2 of between about 5 nm and about 20 nm, such as about 10 nm. Furthermore, the inner spacers **403** may be spaced apart by a first spacing

S1 of between about 3 nm and about 10 nm, such as about 5 nm. However, any suitable widths, heights, and distances may be utilized.

[0056] FIG. 4C further illustrates relative dimensions between the first layers **205**, the second layers **207**, and the inner spacers **403**. According to some embodiments, the second height H2 of the inner spacers **403** is greater than the height of the first layers **205** (e.g., first thickness Th1). In some embodiments, the first spacing S1 between the inner spacers **403** is less than the second thickness Th2 of the second layers **207**. However, any suitable dimensions may be utilized.

[0057] FIGS. 5A-5C illustrates cross-sectional views of forming bottom spacers **501** and source/drain regions **503**, in accordance with some embodiments. In particular, FIG. 5C illustrates a magnified view of a section **505** highlighted in FIG. 5A with a dashed line.

[0058] Once the inner spacers **403** have been formed, the bottom spacers **501** are formed at the bottom of the first openings **401** using a semiconductor material, such as silicon germanium (SiGe), although other suitable materials such as those suitable for forming the first layers **205** may also be utilized. According to some embodiments, the bottom spacers **501** are formed using SiGe and are epitaxially grown at the bottom of the first openings **401** using a deposition technique such as epitaxial growth, vapor-phase epitaxy (VPE), molecular beam epitaxy (MBE). However, any other deposition process suitable for forming the first layers **205** may also be used.

[0059] Once deposited, an oxidation process may be performed to increase a density, and/or a dielectric constant of the bottom spacers **501**. As such, the bottom spacers **501** may be formed as silicon germanium oxide (e.g., SiGeOx, SiGeON, SiGeOCN, or the like). After the oxidation process, an etching process (e.g., wet etch) may be performed to remove any oxide formed along sidewalls of the first openings **401** and/or to recess the bottom spacers **501** to a desired height. According to some embodiments, the bottom spacers **501** are formed to a third height H3 of between about 3 nm and about 30 nm, such as about 20 nm. However, any suitable height may be used.

[0060] Once the bottom spacers **501** have been formed to the desired height, a post placement anneal process may be performed to remove the germanium (Ge) from the material of the bottom spacers **501**. According to some embodiments, the anneal process may comprise one or more anneal processes (e.g., a steam anneal, a high temperature anneal, combinations, or the like) can be performed in a furnace or in a rapid thermal processing (RTP) chamber. According to some embodiments, the post placement anneal process comprises a dry anneal using, for example, nitrogen (N₂) at a process temperature in a range from between about 500° C. and about 700° C., such as about 600° C. for a duration of between about 30 minutes and about 180 minutes, such as about 60 min. However, any suitable oxygen sources, process temperatures, and process times may be utilized. Once the post placement anneal process has completed, the germanium has been removed and the bottom spacers **501** are transformed into a dielectric material such as, silicon nitride (SiN), oxide (OX), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon carbonitride (SiCN), or the like).

[0061] FIG. 5C further illustrates a spacer interface **507** between the bottommost spacer of the inner spacers **403** and

the bottom spacer **501**. According to some embodiments, the spacer interface **507** has a first length **L1** that is between about 3 nm and about 15 nm, such as about 5 nm. However, any suitable length may be used for the spacer interface **507**.

[0062] In some other embodiments, the bottom spacers **501** may be formed by depositing a second spacer material such as SiON using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), plasma enhanced chemical vapor deposition (PECVD), or the like. However, any suitable spacer material and deposition process may be used such as those materials and processes suitable for forming the top spacers **309**. According to some embodiments, the second spacer material is different from the material of the top spacers **309** and the inner spacers **403** and is deposited to fill and/or overfill the first openings **401**.

[0063] Once deposited, the second spacer material may be etched in order to recess the bottom spacers **501** to the desired third height **H3** and to expose the sidewalls in the first openings **401** above the bottom spacers **501**. According to some embodiments, the second spacer material may be etched using an anisotropic etching process (e.g., a dry etching process) such as a reactive ion etching (RIE) process, an isotropic etching process (e.g., a wet etching process), combination thereof, or the like. The etchants used to recess the bottom spacers **501** are selective to the second spacer material without significantly removing the materials of the top spacers **309**, the second layers **207**, and the inner spacers **403**.

[0064] Once the bottom spacers **501** have been formed, the source/drain regions **503** may be formed over the bottom spacers **501**. The source/drain regions **503** may be formed using a growth process such as a selective epitaxial process with a semiconductor material suitable for the device desired to be formed. For example, in an embodiment in which the source/drain regions **503** are utilized to form an NMOS device, the source/drain regions **503** may be a semiconductor material such as silicon, silicon phosphorous, silicon carbon phosphorous, combinations, of these, or the like.

[0065] The epitaxial growth process may use precursors such as silane, dichlorosilane, germane, and the like, and may continue for between about 5 minutes and about 120 minutes, such as about 30 minutes. According to some embodiments, the source/drain regions **503** are formed to a fourth height **H4** of between about 30 nm and about 90 nm, such as about 50 nm. However, any suitable heights and/or suitable depths may be used.

[0066] Once the source/drain regions **503** are formed, dopants may be implanted into the source/drain regions **503** by implanting appropriate dopants to complement the dopants within the remainder of the first device region. For example, n-type dopants such as phosphorous (P), carbon (C), arsenic (As), silicon (Si), antimony (Sb), or the like, and combinations thereof (e.g., SiP, SiC, SiPC, SiAs, Si, Sb, etc.) may be implanted to form NMOSFET devices. These dopants may be implanted using the dummy gate stacks **301** and the top spacers **309** as masks.

[0067] In another embodiment, the dopants of the source/drain regions **503** may be placed during the growth of the source/drain regions **503**. For example, phosphorous may be placed in situ as the source/drain regions **503** are being formed. Any suitable process for placing the dopants within the source/drain regions **503** may be utilized, and all such processes are fully intended to be included within the scope of the embodiments. Furthermore, an anneal process may be

performed to activate the dopants within the source/drain regions **503**. During the anneal process, dopants of the source/drain regions **503** may laterally diffuse into the second layers **207** at the interfaces between the second layers **207** and the source/drain regions **503**. As such, lightly doped drain (LDD) regions may be formed within the second layers **207**.

[0068] FIGS. 6A-6B illustrates cross-sectional views of forming of a contact etch stop layer **601** and an interlayer dielectric layer **603**, in accordance with some embodiments. FIGS. 6A-6B further illustrate the removal of the first hard mask **305** and the second hard mask **307** and planarization of the contact etch stop layer **601** and the interlayer dielectric layer **603** with the dummy gate electrode **303** and top spacers **309**.

[0069] The contact etch stop layer **601** is formed over the structure illustrated in FIGS. 5A-5C, and an interlayer dielectric layer **603** is formed over the contact etch stop layer **601**. The contact etch stop layer **601** functions as an etch stop layer in a subsequent etching process, and may comprise a suitable material such as silicon nitride (SiN), silicon oxynitride (SiON), silicon carbon nitride (SiCN), combinations thereof, or the like, and may be formed by a suitable formation method such as chemical vapor deposition (CVD), physical vapor deposition (PVD), combinations thereof, or the like.

[0070] The interlayer dielectric layer **603** may comprise a material such as silicon dioxide, a low-k dielectric material (e.g., a material having a dielectric constant lower than silicon dioxide), such as silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), organosilicate glasses (OSG), SiOxCy, Spin-On-Glass, Spin-On-Polymers, silicon carbon material, a compound thereof, a composite thereof, the like, or a combination thereof, although any suitable dielectrics may be used. The interlayer dielectric layer **603** may be formed using a process such as plasma enhanced chemical vapor deposition (PECVD), although other processes, such as low pressure chemical vapor deposition (LPCVD), may also be used.

[0071] Once formed, the interlayer dielectric layer **603** and the contact etch stop layer **601** may be planarized with the dummy gate electrode **303** and top spacers **309** using a planarization process such as chemical mechanical planarization (CMP). However, any suitable planarization process may be utilized. Furthermore, the first hard mask **305** and the second hard mask **307** may be removed during the planarization process. According to some embodiments, one or more etching processes and/or the chemical mechanical planarization (CMP) may be utilized to remove the first hard mask **305** and the second hard mask **307**. As such, the dummy gate electrode **303** is exposed after the removal of the first hard mask **305**.

[0072] FIGS. 7A-7B illustrates cross-sectional views of removing the dummy gate electrode **303** and the dummy gate dielectric **211**. FIGS. 7A-7B further illustrate a wire-release process to form nanostructures **701** from the second layers **207**, in accordance with some embodiments. FIGS. 7A-7B further illustrate the formation of a gate dielectric **703** over the nanostructures **701**, according to some embodiments.

[0073] Once exposed, the dummy gate electrode **303** may be removed in order to expose the underlying dummy gate

dielectric **211**. In an embodiment the dummy gate electrode **303** is removed using, e.g., one or more wet or dry etching process that utilizes etchants that are selective to the material of the dummy gate electrode **303**. However, any suitable removal process may be utilized.

[0074] Once the dummy gate dielectric **211** has been exposed, the dummy gate dielectric **211** may be removed. In an embodiment the dummy gate dielectric **211** may be removed using, e.g., a wet etching process, although any suitable etching process may be utilized.

[0075] FIGS. 7A-7B further show that, once the dummy gate dielectric **211** has been removed (which also exposes the sides of the first layers **205**), the first layers **205** may be removed from between the substrate **201** and from between the second layers **207** in a wire release process step. The wire release process step may also be referred to as a sheet release process step, a sheet formation process step, a nanosheet formation process step or a wire formation process step. In an embodiment the first layers **205** may be removed using a wet etching process that selectively removes the material of the first layers **205** (e.g., silicon germanium (SiGe)) without significantly removing the material of the substrate **201** and the material of the second layers **207** (e.g., silicon (Si)). However, any suitable removal process may be utilized.

[0076] For example, in an embodiment, an etchant such as a high temperature HCl may be used to selectively remove the material of the first layers **205** (e.g., SiGe) without substantially removing the material of the substrate **201** and/or the material of the second layers **207** (e.g., Si). Additionally, the wet etching process may be performed at a temperature of between about 400° C. and about 600° C., such as about 560° C., and for a time of between about 100 seconds and about 600 seconds, such as about 300 seconds. However, any suitable etchant, process parameters, and time can be utilized.

[0077] By removing the material of the first layers **205**, the sides of the second layers **207** (reabeled nanostructures **701** in FIGS. 7A-7B) are exposed. The nanostructures **701** are separated from each other by the inner spacers **403**. According to some embodiments, the nanostructures **701** are separated by a spacing of between about 5 nm and about 15 nm, such as about 10 nm. The nanostructures **701** comprise the channel regions between opposite ones of the source/drain regions **503** and have a channel length CL1 of between about 5 nm and about 180 nm, such as about 10 nm and a channel width CW1 of between about 8 nm and about 100 nm, such as about 30 nm. In an embodiment the nanostructures **701** are formed to have the same thicknesses as the original thicknesses of the second layers **207** such as, of between about 3 nm and about 15 nm, such as about 8 nm, although the etching processes may also be utilized to reduce the thicknesses.

[0078] In some embodiments, the wire release step may include an optional step for the partial removal of the material of the second layers **207** (e.g., by over etching) during removal of the first layers **205**. As such, the thicknesses of the nanostructures **701** are formed to have reduced thicknesses as compared to the original thickness of the second layers **207**. As such, the nanostructures **701** may have third thicknesses Th3 that are less than the thicknesses of the original second layers **207** (e.g., second thicknesses Th2). Furthermore, during such a partial removal, the top spacers **309**, the inner spacers **403** may serve to protect

adjacent material of the second layers **207** from being removed. As such, the thicknesses of the nanostructures **701** at distal end portions of the nanostructures **701** are protected during the wire release step and are not further reduced.

[0079] Additionally, although FIGS. 7A to 7C illustrate the formation of three of the nanostructures **701**, any suitable number of the nanostructures **701** may be formed from the nanosheets provided in the multi-layer stack **203**. For example, the multi-layer stack **203** may be formed to include any suitable number of the first layers **205** and any suitable number of the second layers **207**. As such, a multi-layer stack **203** comprising fewer first layers **205** and fewer second layers **207**, after removal of the first layers **205**, forms one or two of the nanostructures **701**. Whereas, a multi-layer stack **203** comprising many of the first layers **205** and many of the second layers **207**, after removal of the first layers **205**, forms four or more of the nanostructures **701**.

[0080] FIGS. 7A-7B further illustrate the formation of the gate dielectric **703** over the nanostructures **701**, according to some embodiments. In an embodiment the gate dielectric **703** comprises a high-k material (e.g., $K \geq 9$) such as Ta₂O₅, Al₂O₃, Hf oxides, Ta oxides, Ti oxides, Zr oxides, Al oxides, La oxides (e.g., HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, LaO, ZrO, TiO), combinations of these, or the like, deposited through a process such as atomic layer deposition, chemical vapor deposition, or the like. In some embodiments, the gate dielectric **703** comprises a nitrogen doped oxide dielectric that is initially formed prior to forming a metal content high-K (e.g., $K \text{ value} > 13$) dielectric material. The gate dielectric **703** may be deposited to a thickness of between about 1 nm and about 3 nm, although any suitable material and thickness may be utilized. As illustrated, the gate dielectric **703** wraps around the nanostructures **701**, thus forming channel regions between the source/drain regions **503**.

[0081] FIGS. 8A and 8B illustrate cross-sectional views of forming gate electrodes **107** and gate caps **801**, in accordance with some embodiments. Once the gate dielectric **703** has been formed, the gate electrodes **107** are formed to surround the nanostructures **701**. In some embodiments, the gate electrodes **107** are formed using multiple layers, each layer deposited sequentially adjacent to each other using a highly conformal deposition process such as atomic layer deposition, although any suitable deposition process may be utilized. According to some embodiments, the gate electrodes **107** may comprise a capping layer, a barrier layer, an n-metal work function layer, a p-metal work function layer, and a fill material.

[0082] The capping layer may be formed adjacent to the gate dielectric **703** and may be formed from a metallic material such as TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. The metallic material may be deposited using a deposition process such as atomic layer deposition, chemical vapor deposition, or the like, although any suitable deposition process may be used.

[0083] The barrier layer may be formed adjacent the capping layer, and may be formed of a material different from the capping layer. For example, the barrier layer may be formed of a material such as one or more layers of a

metallic material such as TiN, TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. The barrier layer may be deposited using a deposition process such as atomic layer deposition, chemical vapor deposition, or the like, although any suitable deposition process may be used.

[0084] The n-metal work function layer may be formed adjacent to the barrier layer. In an embodiment the n-metal work function layer is a material such as W, Cu, AlCu, TiAlC, TiAlN, TiAl, Pt, Ti, TiN, Ta, TaN, Co, Ni, Ag, Al, TaAl, TaAlC, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. For example, the first n-metal work function layer may be deposited utilizing an atomic layer deposition (ALD) process, CVD process, or the like. However, any suitable materials and processes may be utilized to form the n-metal work function layer.

[0085] The p-metal work function layer may be formed adjacent to the n-metal work function layer. In an embodiment, the first p-metal work function layer may be formed from a metallic material such as W, Al, Cu, TiN, Ti, TiAlN, TiAl, Pt, Ta, TaN, Co, Ni, TaC, TaCN, TaSiN, TaSi₂, NiSi₂, Mn, Zr, ZrSi₂, TaN, Ru, AlCu, Mo, MoSi₂, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. Additionally, the p-metal work function layer may be deposited using a deposition process such as atomic layer deposition, chemical vapor deposition, or the like, although any suitable deposition process may be used.

[0086] Once the p-metal work function layer has been formed, the fill material is deposited to fill a remainder of the opening. In an embodiment the fill material may be a material such as tungsten, Al, Cu, AlCu, W, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ta, TaN, Co, Ni, combinations of these, or the like, and may be formed using a deposition process such as plating, chemical vapor deposition, atomic layer deposition, physical vapor deposition, combinations of these, or the like. However, any suitable material may be utilized.

[0087] Once the openings left behind by the removal of the dummy gate electrode **303** have been filled, the materials of the gate electrode **107** and the gate dielectric **703** may be planarized in order to remove any material that is outside of the openings left behind by the removal of the dummy gate electrode **303**. In a particular embodiment the removal may be performed using a planarization process such as chemical mechanical polishing, although any suitable planarization and removal process may be utilized. According to some embodiments, the gate electrodes may be formed to a fifth length **L5** of between about 8 nm and about 30 nm. However, any suitable length may be used.

[0088] Once formed, the gate electrodes **107** may be recessed below the planarized surfaces of the interlayer dielectric layer **603**. According to some embodiments, the gate electrodes **107** may be recessed using an etching process such as a wet etch, a dry etch, combinations, or the like. Once recessed, the height of the gate electrodes **107** above a topmost one of the nanostructure **701** is a fifth height

H5. According to some embodiments, the fifth height **H5** is between about 8 nm and about 30 nm. However, any suitable height may be used.

[0089] The gate caps **801** may be formed by initially depositing a dielectric material over the gate electrodes **107** to fill and/or overfill the recesses. In some embodiments, the gate caps **801** are formed using a dielectric material such as a silicon nitride (SiN), oxide (OX), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon carbonitride (SiCN), or the like. According to some embodiments, the gate caps **801** are formed using a metal oxide of materials such as zirconium (Zr), hafnium (Hf), aluminium (Al), or the like. Furthermore, the gate caps **801** may be formed using a suitable deposition process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), combinations of these, or the like. However, any suitable materials and deposition processes may be utilized. Once deposited, the gate caps **801** may be planarized with the interlayer dielectric layer **603** using a planarization process such as a chemical mechanical polishing process. Once planarized, the gate caps **801** have a sixth height **H6** of between about 10 nm and about 30 nm. However, any suitable height may be used.

[0090] FIGS. 9A and 9B illustrate cross-sectional views of forming second openings **901** in a cut-metal gate process, in accordance with some embodiments. Once the gate caps **801** have been planarized with the interlayer dielectric layer **603**, a masking layer **903** may be deposited over the planar surfaces of the gate caps **801**, the interlayer dielectric layer **603**, the contact etch stop layer **601**, the top spacers **309**, and the gate dielectric **703**. Once deposited, the masking layer **903** is patterned to expose the underlying materials including the gate caps **801** and the interlayer dielectric layer **603** in desired locations of the cut-metal gate structures **109** that are to be formed.

[0091] Once patterned, the masking layer **903** is used as an etching mask to etch the underlying materials to form the second openings **901** (e.g., trenches, recesses, channels or the like). In the etching process, the materials of the gate caps **801** and the gate electrodes **107** are etched using an anisotropic etching process, and may stop on the top surface of the gate dielectric **703** or the top surface of the isolation regions **209**. The second openings **901** may be formed between adjacent ones of the fins **105** and may be formed to cut through one or more gate electrodes **107**. According to some embodiments, two of the second openings **901** are formed to cut through two adjacent ones of the gate electrodes **107** and are located on opposite sides of one of the fins **105**, as shown in FIG. 1. According to some embodiments, the second openings **901** are located between top spacers **309** of two neighboring devices. Once the second openings **901** have been formed, the masking layer **903** is removed.

[0092] FIGS. 10A-10B illustrates cross-sectional views of forming cut-metal gate structures **109**, in accordance with some embodiments. Once the second openings **901** have been formed, the cut-metal gate structures **109** are formed by initially depositing a dielectric material to fill and overfill the second openings **901**. In accordance with some embodiments, the cut-metal gate structures **109** are formed using any dielectric material and deposition process suitable for forming the gate caps **801**. In some embodiments, the dielectric material used to form the cut-metal gate structures **109** is the same as the dielectric material used to form the

gate caps **801**, although the dielectric materials may be different. For example, in embodiments where the gate caps **801** are formed using silicon nitride (SiN), the cut-metal gate structures **109** may also be formed using silicon nitride (SiN) in a deposition process such as Atomic Layer Deposition (ALD). However, any suitable dielectric materials and deposition processes may be used. According to some embodiments, the cut-metal gate structures **109** are formed to a third width **W3** of between about 5 nm and about 50 nm, such as about 10 nm. However, any suitable widths may be used.

[0093] The cut-metal gate structures **109** divide the two gate electrodes **107** which are relatively long into a plurality of gate electrodes **107** which are relatively short and isolate the plurality of gate electrodes **107** from one another. Furthermore, the excess dielectric material of the cut-metal gate structures **109** outside of the second openings **901** may be retained and used as a masking layer in the Continuous Poly On Diffusion Edge (CPODE) process. As such, the cut-metal gate structures **109** are highlighted with a dashed line in FIG. **10B** with the excess dielectric material remaining outside of the second openings **901**.

[0094] FIGS. **11A-11B** illustrates cross-sectional views of forming a third opening **1003** in an initial step of forming a Continuous Poly On Diffusion Edge (CPODE) structure **111**, in accordance with some embodiments. The CPODE structure **111** may also be referred to herein as an isolation structure, a cut-poly structure or a cut-PODE structure and is discussed in greater detail with the following figures.

[0095] Once the cut-metal gate structures **109** have been formed, a photo resist may be formed over the excess dielectric material and openings may be formed in the photo resist in a desired location of the CPODE structure **111** to be formed. The opening in the photo resist is formed to reveal a portion of the gate cap **801** between the two cut-metal gate structures **109**. Furthermore, edge portions of the cut-metal gate structures **109** may also be exposed through the openings in the photo resist in order to provide some process margin for the CPODE structure **111**. According to some embodiments, the width of the exposed edge portions may be a fourth width **W4** of between about 3 nm and about 25 nm, such as about 5 nm. However, any suitable width may be used.

[0096] The photo resist is then used as an etching mask to etch the excess dielectric material, edge portions of the cut-metal gate structures **109**, the gate cap **801**, and the gate electrode **107**, so that the third opening **1003** (e.g., trenches, recesses, channels or the like) is formed in the desired location of the CPODE structure **111**. According to some embodiments, the etching process may stop on the gate dielectric **703**. As a result, the gate dielectric **703** and the nanostructures **701** remain at the bottom of the third opening **1003**. According to some embodiments, the etching process used to form the third opening **1003** may be an isotropic etching process (e.g., a wet etching process) using etchants that stop on the gate dielectric **703**. However, other suitable etching process including anisotropic etching processes (e.g., a dry etching processes or reactive ion etching (RIE) processes), combinations of isotropic and anisotropic etches, or the like may also be used. According to some embodiments, the third opening **1003** is formed to a fifth width **W5** of between about 20 nm and about 200 nm, such as about 70 nm. However, any suitable width may be used.

[0097] FIGS. **12A-12B** illustrates cross-sectional views of further etching processes in an intermediate step of performing the CPODE process, in accordance with some embodiments. Once the gate dielectric **703** has been exposed, another etching process is performed to remove the materials of the gate dielectric **703** within the third opening **1003** and to expose the nanostructures **701**, the fin **105**, and isolation regions **209** within the third opening **1003**. According to some embodiments, a wet etch, a dry etch, combinations, or the like may be used to remove the material of the gate dielectric **703** without substantially removing the materials of the top spacers **309** and the inner spacers **403** along vertical sidewalls of the third opening **1003**. However, any suitable etching process may be used.

[0098] Once the nanostructures **701** and a portion of the fin **105** protruding above the isolation regions **209** are exposed, a further etching process may be used to remove the materials of the nanostructures **701** and to form the recess within the fin **105**. According to some embodiments, an etching process such as a wet etch, a dry etch, combinations, or the like may be used to remove these materials without substantially removing the materials of the top spacers **309**, the inner spacers **403**, and the isolation regions **209** the third opening **1003**. According to an embodiment, the portion of the fin **105** protruding above the isolation regions **209** may be recessed a second depth **D2** of between about 2 nm and about 20 nm, such as about 10 nm. However, any suitable depth may be used. According to some embodiments, a first ratio **R1** of the second depth **D2** to the third height **H3** of the bottom spacer **501** may be between about 0.05:1 and about 1:1. Once the nanostructures **701** have been removed and the portion of the fin **105** protruding above the isolation regions **209** has been recessed, the photo resist may be removed, for example, via an ashing process.

[0099] FIGS. **13A-13B** illustrates cross-sectional views of forming the CPODE structure **111**, in accordance with some embodiments. The CPODE structure **111** may be formed by depositing a dielectric material to fill and/or overfill the third opening **1003**. The CPODE structure **111** may be formed using a dielectric material such as a silicon nitride (SiN), oxide (OX), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon carbonitride (SiCN), or the like. According to some embodiments, the CPODE structure **111** is formed using a metal oxide of materials such as zirconium (Zr), hafnium (Hf), aluminum (Al), or the like. Furthermore, the CPODE structure **111** may be formed using a suitable deposition process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), combinations of these, or the like. The CPODE structure **111** may be formed using any of the dielectric materials and processes suitable for forming the cut-metal gate structures **109**. According to some embodiments, the dielectric material used to form the CPODE structure **111** is the same dielectric material used to form the cut-metal gate structures **109**, although the dielectric materials may also be different. For example, in an embodiment where the cut-metal gate structures **109** are formed using silicon nitride (SiN), the CPODE structure **111** may be formed using silicon nitride (SiN) via a deposition process such as, chemical vapor deposition (CVD), atomic layer deposition (ALD), sputtering, or the like.

[0100] Once formed, the excess material of the CPODE structure **111** outside of the third opening **1003** may be removed by a chemical mechanical planarization (CMP)

process. According to some embodiments, the chemical mechanical planarization (CMP) process may be continued to planarize the surfaces of the CPODE structure 111 with the gate electrode 107 and the interlayer dielectric layer 603. Once planarized, the CPODE structure 111 over the fin 105 has a seventh height H7 of between about 55 nm and about 140 nm, such as about 70 nm. However, any suitable height may be used. The gate caps 801 have an eighth height H8 of between about 10 nm and about 30 nm, such as about 15 nm. However, any suitable height may be used. Furthermore, the cut-metal gate structures 109 have a ninth height H9 of between about 50 nm and about 120 nm, such as about 60 nm. However, any suitable height may be used.

[0101] FIGS. 14A and 14B illustrate cross-sectional views of forming source/drain contacts 1401 to the source/drain regions 503, in accordance with some embodiments. FIGS. 14A-14B further illustrate the formation of source/drain vias 1407 and gate vias 1409, in accordance with some embodiments.

[0102] Once the CPODE structure 111 has been formed, silicide contacts 1403 and source/drain contacts 1401 may be formed through the interlayer dielectric layer 603 to make electrical connection to the source/drain regions 503. In an embodiment the silicide contacts 1403 and the source/drain contacts 1401 may be formed by initially forming openings through the interlayer dielectric layer 603 in order to expose the source/drain regions 503. The openings may be formed using, e.g., a suitable photolithographic masking and etching process.

[0103] The silicide contacts 1403 may comprise titanium, nickel, cobalt, or erbium in order to reduce the Schottky barrier height of the source/drain contacts 1401. However, other metals, such as platinum, palladium, and the like, may also be used. The silicidation may be performed by blanket deposition of an appropriate metal layer, followed by an annealing step which causes the metal to react with the underlying exposed silicon of the source/drain regions 503. Un-reacted metal is then removed, such as with a selective etch process. The thickness of the silicide contacts 1403 may be between about 5 nm and about 50 nm. However, any suitable thickness may be used.

[0104] In an embodiment the source/drain contacts 1401 may be a conductive material such as Al, Cu, W, Co, Ti, Ta, Ru, TiN, TiAl, TiAlN, TaN, TaC, NiSi, CoSi, combinations of these, or the like, although any suitable material may be deposited into the openings using a deposition process such as sputtering, chemical vapor deposition, electroplating, electroless plating, or the like, to fill and/or overfill the openings. Once filled or overfilled, any deposited material outside of the openings may be removed using a planarization process such as chemical mechanical polishing (CMP). However, any suitable material and process of formation may be utilized.

[0105] Once the source/drain contacts 1401 have been formed, a second ILD layer 1405 may be formed by depositing a dielectric material over the planar surfaces of the interlayer dielectric layer 603, contact etch stop layer 601, the top spacers 309, the gate dielectric 703, the gate caps 801, the cut-metal gate structures 109 and the CPODE structure 111. The second ILD layer 1405 may be formed and planarized using any of the processes and materials suitable for forming the interlayer dielectric layer 603, as set forth above.

[0106] Once the second ILD layer 1405 has been formed, the source/drain vias 1407 may be formed through the second ILD layer 1405 to provide electrical connectivity to the source/drain regions 503. Furthermore, the gate vias 1409 may be formed through the second ILD layer 1405 and the gate caps 801 to provide electrical connectivity to the gate electrodes 107. Any suitable masking and etching process may be used to form the openings through the second ILD layer 1405 and through the dielectric capping layer 1104. In some embodiments, the openings for the source/drain vias 1407 and the openings for the gate vias 1409 may be patterned together using the same etching process. In other embodiments, the openings for the source/drain vias 1407 and the openings for the gate vias 1409 may be patterned in separate etching processes.

[0107] Once the openings have been formed for the source/drain vias 1407 and the gate vias 1409, conductive material, such as copper, may be deposited to fill and/or overfill the openings using a deposition process such as plating, chemical vapor deposition, sputtering, combinations of these, or the like. Excess material may then be removed using, for example, a planarization process such as chemical mechanical planarization, or the like.

[0108] According to the embodiments disclosed herein, the bottom spacers 501 are formed with the spacer interfaces 507 to the bottommost spacers of the inner spacers 403. The spacer interfaces 507 may have a length of between about 3 nm and about 15 nm. As such, the epitaxial growth of the source/drain regions 503 is ensured to be isolated from the substrate 201. If the length L1 of the spacer interfaces 507 is too short (e.g., less than 3 nm), the epitaxial growth of the source/drain regions 503 may not be formed in isolation from the substrate 201 when taking into consideration an etching depth variation for the source/drain regions 503. Furthermore if the length L1 of the spacer interfaces 507 is too long (e.g., greater than 15 nm) and is greater than the height H2 of the inner spacers 403, the bottom spacers 501 may hinder the current injection from the source/drain regions 507 into the nanostructures 701 during operation. As such, the production yield and/or device performance may be degraded.

[0109] By isolating the source/drain regions 503 from the substrate 201, the CPODE structure 111 may be formed with a shallow recess into the fin 105 without cutting through the isolation regions 209. According to some embodiments, the shallow recess may be recessed to the second depth D2 of between about 2 nm and about 20 nm. The first ratio R1 defines a ratio of the second depth D2 of the shallow recess used to form the CPODE structure 111 to the third height H3 of the bottom spacer 501. As such, the first ratio R1 can be used to determine the second depth D2 of the shallow recess to be used in forming the CPODE structure 111 for different technology nodes. According to some embodiments, the first ratio R1 may be between about 0.05:1 and about 1:1. By forming the CPODE structure 111 with the shallow recess with the first ratio R1 being roughly less than 1:1 allows for a large source/drain damage window and allows for accurate control of the gate height. With the large source/drain damage window, the CPODE structure 111 may be formed without damaging the source/drain regions 503 even when applied to N5 technologies and beyond. Furthermore, due to the accurate control of the gate height, little to no GH loss is experienced during formation of the CPODE structure 111. As such, forming the CPODE structure 111 using the

second depth D2 for the shallow recess depth may be applied to N3 technologies and beyond. Furthermore, by forming the CPODE structure 111 using the first ratio R1 of at least 0.05:1 ensures efficient cut off of the nanostructures 701 for isolating the source/drain regions 503 from the substrate 201 during device operation. As such, production yield is increased and/or device performance is improved.

[0110] According to an embodiment, a method includes: forming a fin in a multilayer stack, the fin including a substrate and at least one first layer over the substrate; forming a gate electrode over the fin; etching an opening in the fin adjacent the gate electrode; forming a recess along a sidewall of the opening, the recess being formed in the at least one first layer; depositing an spacer material in the recess; forming a bottom spacer at a bottom of the opening, the bottom spacer forming a first interface with the spacer material; forming a source/drain region over the bottom spacer; and forming a dielectric structure through the gate electrode and into the substrate of the fin, a bottom of the dielectric structure being above a bottom of the bottom spacer. In an embodiment of the method, the spacer material within the recess has a thickness larger than the at least one first layer. In an embodiment of the method, the bottom spacer is formed to a height of between about 3 nm and about 30 nm. In an embodiment of the method, a length of the first interface is between about 3 nm and about 15 nm. In an embodiment, the method further includes forming a cut gate dielectric prior to the forming the dielectric structure. In an embodiment, the method further includes forming isolation regions on opposite sides of the fin, wherein the forming the gate electrode includes forming the gate electrode over the isolation regions and wherein a level of the bottom of the dielectric structure is between upper surfaces and bottom surfaces of the isolation regions. In an embodiment of the method, the level of the bottom of the dielectric structure extends into the substrate a distance of between about 2 nm and about 20 nm.

[0111] According to another embodiment, a method includes: etching a first opening through a gate electrode, a gate dielectric being exposed within the first opening; removing the gate dielectric within the first opening, wherein after the removing the gate dielectric a portion of a substrate, a stack of nanowires, and isolation regions are exposed; performing an etching process to remove the stack of nanowires within the first opening and to recess the portion of the substrate to a first level between an upper surface of the isolation regions and a bottom surface of the isolation regions; and depositing a dielectric material in the first opening, the dielectric material isolating a first portion of the gate electrode from a second portion of the gate electrode. In an embodiment of the method, the first level is at most 20 nm below the upper surface of the isolation regions. In an embodiment of the method, the first level is at least 2 nm below the upper surface of the isolation regions. In an embodiment, the method further includes forming a bottom spacer adjacent the substrate prior to the etching the first opening through the gate electrode; and forming a source/drain region over the bottom spacer. In an embodiment of the method, a bottom of the bottom spacer is below the first level. In an embodiment, the method further includes forming an inner spacer in a second stack of nanostructures, the inner spacer having a first interface with

the bottom spacer. In an embodiment of the method, the first interface has a length of between about 3 nm and about 15 nm.

[0112] According to an embodiment, a semiconductor device includes: a stack of nanostructures over a first semiconductor fin, the first semiconductor fin being located over a substrate; an isolation region extending between the first semiconductor fin and a second semiconductor fin located over the substrate, the isolation region having a first surface facing away from the substrate, the first surface being located a first distance away from the substrate; a gate electrode surrounding each of the nanostructures within the stack of nanostructures; and a dielectric material adjacent to the gate electrode, the dielectric material having a first portion located adjacent to the second semiconductor fin, the first portion being located a second distance away from the substrate, the second distance being less than the first distance. In an embodiment of the semiconductor device, the second distance is at most 20 nm less than the first distance. In an embodiment, the semiconductor device further includes: a first source/drain region adjacent the stack of nanostructures; a first bottom spacer isolating the first source/drain region from the substrate; an inner spacer adjacent the first bottom spacer and separating the first bottom spacer from the dielectric material. In an embodiment of the semiconductor device, the inner spacer includes a first interface with the first bottom spacer, the first interface having a distance of between about 3 nm and about 15 nm. In an embodiment of the semiconductor device, the inner spacer has a first height of between about 5 nm and about 20 nm. In an embodiment of the semiconductor device, the first height is greater than a second height of one of the nanostructures within the stack of nanostructures.

[0113] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device comprising:

- a first nanostructure and a second nanostructure overlying a fin, the fin protruding from a substrate;
- an isolation region adjacent the fin;
- a first inner spacer located between the first nanostructure and the semiconductor substrate;
- a second inner spacer located between the second nanostructure and the semiconductor substrate;
- a first spacer material in contact with the first inner spacer and the second inner spacer;
- a first source/drain region over an upper surface of the first spacer material, wherein the first source/drain region contacts each of the first spacer material, the first inner spacer, and the second inner spacer;
- a first gate electrode surrounding the first nanostructure;

a second gate electrode adjacent the first gate electrode and having a longitudinal axis parallel to a longitudinal axis of the first gate electrode; and

a dielectric structure extending through the second gate electrode and into the fin, a bottommost surface of the dielectric structure being higher than a bottom of the first spacer material, wherein the second inner spacer is between the dielectric structure and the first spacer material, wherein the second nanostructure is between the dielectric structure and the first source/drain region.

2. The semiconductor device of claim 1, wherein the dielectric structure comprises:

- a first layer having a bottom surface on a top surface of the isolation region;
- a second layer having a bottom surface on the top surface of the isolation region; and
- a third layer between the first layer and the second layer, the third layer extending into the fin.

3. The semiconductor device of claim 2, wherein the third layer partially overlaps the first layer and the second layer.

4. The semiconductor device of claim 1, wherein the first spacer material extends lower than the dielectric structure.

5. The semiconductor device of claim 1, wherein a height of the first nanostructure adjacent the first source/drain region is less than a height of the first nanostructure between portions of the first gate electrode.

6. The semiconductor device of claim 1, wherein the first spacer material overlaps the first inner spacer by a distance in a range of 3 nm to 15 nm.

7. The semiconductor device of claim 1, wherein the dielectric structure extends into the fin by a distance in a range of 2 nm to 20 nm.

8. A semiconductor device comprising:

- a first gate structure;
- a second gate structure, the first gate structure and the second gate structure being aligned along a first longitudinal axis;
- a third gate structure, the third gate structure having a second longitudinal axis, the second longitudinal axis being parallel to the first longitudinal axis;
- a first gate cut structure between the first gate structure and the second gate structure;
- a second gate cut structure between the first gate structure and the second gate structure, the first gate cut structure being spaced apart from the second gate cut structure;
- a fin;
- an isolation region adjacent the fin;
- a fin divider extending into the fin, the fin divider extending from the first gate cut structure to the second gate cut structure, wherein the fin divider overlaps the first gate cut structure and the second gate cut structure, the fin divider extending lower than an upper surface of the isolation region;
- a bottom source/drain spacer between the third gate structure and the fin divider, wherein the bottom source/drain spacer extends above an upper surface of the fin; and
- a source/drain region over the bottom source/drain spacer adjacent the third gate structure.

9. The semiconductor device of claim 8, further comprising:

- a first nanostructure, wherein the third gate structure extends along sidewalls of the first nanostructure; and

- a first inner spacer between the first nanostructure and the fin, wherein the first inner spacer contacts the bottom source/drain spacer.

10. The semiconductor device of claim 9, wherein the source/drain region extends lower than an upper surface of the first inner spacer.

11. The semiconductor device of claim 9, wherein the source/drain region contacts the first inner spacer.

12. The semiconductor device of claim 8, wherein an upper surface of the first gate cut structure and an upper surface of the fin divider are level.

13. The semiconductor device of claim 8, wherein an upper width of the first gate cut structure is less than a bottom width of the first gate cut structure.

14. The semiconductor device of claim 8, wherein the bottom source/drain spacer extends further into the fin than the fin divider.

15. A semiconductor device comprising:

- a first stack of nanostructures;
- a gate structure along sidewalls of the first stack of nanostructures;
- a first source/drain region adjacent the first stack of nanostructures;
- a bottom spacer under the first source/drain region;
- a first inner spacer below a bottommost nanostructure of the first stack of nanostructures, the first inner spacer contacting the bottommost nanostructure, the bottom spacer, and the first source/drain region; and
- a dielectric structure adjacent to the first source/drain region, the first source/drain region being between the dielectric structure and the gate structure, a bottom surface of the bottom spacer being lower than a bottom surface of the dielectric structure.

16. The semiconductor device of claim 15, wherein the dielectric structure comprises:

- a first layer;
- a second layer; and
- a third layer between the first layer and the second layer, the third layer extending lower than the first layer and the second layer, wherein an upper surface of the third layer is level with an upper surface of the first layer and an upper surface of the second layer.

17. The semiconductor device of claim 16, wherein the first stack of nanostructures is over a semiconductor fin, wherein the dielectric structure extends into the semiconductor fin.

18. The semiconductor device of claim 17, further comprising:

- an isolation region adjacent the semiconductor fin, wherein the first layer and the second layer are over an upper surface of the isolation region, wherein a bottom surface of the third layer extends below the upper surface of the isolation region.

19. The semiconductor device of claim 15, further comprising:

- a second stack of nanostructures between the dielectric structure and the first source/drain region, wherein the second stack of nanostructures extends from the first source/drain region to the dielectric structure.

20. The semiconductor device of claim 15, wherein the first source/drain region contacts the first inner spacer.